

DEVELOPMENT OF A NEURAL NETWORK-BASED CONTROL SYSTEM FOR
TRAJECTORY TRACKING, BERTHING AND UNBERTHING MISSIONS

SUPPORTING DOCUMENTATION.

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ABSTRACT

The benefits of autonomous ship control have been on the spotlight in recent times. In very busy shipping ports where safety and reliability is paramount, the use of artificial intelligence in the areas of automatic berthing, unberthing and trajectory tracking of surface vessels has proved to be of immense benefits over the conventional use of tugs, lines and manual manoeuvring. These conventional systems are less reliable and in several cases of failure, have led to the loss of lives, grounding and damage to ships, quays and jetties. To solve these problems, the author of this paper presents the development of a neural network-based control algorithm for trajectory tracking, berthing and unberthing mission for surface vessels. The author proposes the use of multilayer perceptron with the backpropagation learning algorithm in the development of an ANN-based autopilot, speed controller and position controller for a surface vessel. To effectively demonstrate the efficiency, reliability and faster calculation of the artificial neural network-based controllers in surface vessels, the conventional PID control law is proposed as a basic design for the same set of controllers and a comparison study is made between the artificial neural network-based controllers and the PID-based controllers. The results of the simulation studies reveal that the artificial neural network-based controllers have a better performance compared to the PID-based controllers in trajectory tracking, berthing and unberthing mission of a ship. Overall, the neural network-based control system has a faster response and an ability to adapt quickly to changes in the ship dynamics.

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1.0 INTRODUCTION

BACKGROUND

Autonomous and intelligent ship berthing and unberthing remains on the spotlight as the inefficiencies of the conventional/traditional use of tugs, lines and other manual methods continue to pose problems. Due to the absence of an efficient system of berthing and unberthing, there has been recorded cases where ships ran aground, collided with other ships, damaged or demolished jetties, hit berths, constituted an environment pollution and in worst cases resulted in the loss of lives. Perhaps, these losses and the need for safety becomes a good rationale to implement efficient and intelligent control systems in the ship berthing and unberthing process.

The success of automatic berthing and unberthing in shipping industries will imply,

- An improved operational efficiency in marine transportation and global shipping,
- risk reduction and an improved security in lives and property,
- an improved environmental protection.

This intelligent control system for berthing and unberthing will seek to eliminate the use of tugboats and other less efficient and traditional methods in use. Traditionally berthing and unberthing of vessels requires a captain to manually maneuver a vessel while making decisions based on visual contacts. The heading and speed of berthing and unberthing are solely left at the discretion of the captain regardless of the limitations of this manual method. Poor human judgements, lack of maneuvering information and poor weather condition could pose challenges to an accurate berthing and unberthing process. This rationale prompts the need for an improved intelligent control system for this process

This study seeks to conduct a numerical simulation for development of a neural network-based control algorithm for trajectory tracking/berthing and unberthing of a surface vessel. This desired control system will be based on an algorithm that can make high level decisions to intelligently guide a surface vessel through a berthing and unberthing process over its entire speed envelope. The validity of the achieved results will be confirmed by experiments using a scaled model of a surface ship, Bluefin.

Several control algorithms like the PID (which has a tuning challenge) and the fuzzy logic theorem have been implemented in designing similar vessel control systems, however, this work will exploit the learning capabilities of the neural network to produce an intelligent control system for vessel berthing and unberthing.

PROBLEM STATEMENT

Traditional berthing and unberthing methods for surface ships are very inefficient and unreliable. With traditional ship berthing and unberthing processes, there has been recorded cases where ships ran aground or hit Jetties, quays or wharfs. Such collision could cause casualties, damage to the ship's structures and parts and an extra operational cost. In commercial shipping operations, there are implications in the form of longer ship turnaround times at ports and a loss of revenue to the ship owner or government due to the delayed, poor and traditional berthing and unberthing methods.

HYPOTHESIS

Unlike conventional controllers used in automatic ship control, adopting an Intelligent control algorithm such as the artificial neural network control will yield a control system with an impressive level of performance for a wide range of operating conditions (Simensen et al 1996). The use of artificial neural network controllers in ship control systems offers the advantages of an adaptive system in terms of the ability of the controller to learn and to handle variations in the ship dynamics. Simensen et al (1996) noted that in ship steering control, selected optimal settings of the conventional controller (like the PID-based controllers) vary when there are changes in operating conditions and dynamics of a ship. When there are significant variations in the ship's forward speed, water depth or the cargo weight, Conventional controllers lack an adaptive system behavior to match these possible variations in ship dynamics and as such do not suffice as an efficient ship control method.

Neural networks thus appear to offer solutions to the problem of adaptation and effect of varying ship dynamics to automatic ship control and trajectory tracking. The methodology for this research project will entail,

- i) the use of the conventional PID control law as a basic design and for comparison studies for automatic ship control and to investigate how the ANN control law can be applied for automatic ship control for trajectory tracking, berthing and unberthing missions,
- ii) to realize the use of the ANN-based controller for various ship missions; including trajectory tracking, berthing and unberthing missions,
- iii) to validate the proposed ANN control law by experiments,
- iv) to conduct a comparison studies to verify that the performance of the proposed ANN control law has a better performance than the use of the conventional PID control law for trajectory tracking, berthing and unberthing of surface ships.

PROJECT AIMS AND OBJECTIVES.

The project objective includes the development of an intelligent control system for trajectory tracking, berthing and unberthing of a surface vessel using neural-network-based control algorithm. This research project seeks to provide and validate an efficient and a reliable trajectory tracking/berthing and unberthing system for surface vessels. Bluefin Model-scale ship will be used for the proposed project work.

This project aims at addressing the current challenges of berthing and unberthing of surface vessels. It aims at developing an efficient, improved and an automated berthing and unberthing system for surface vessels.

This research project will seek to employ the advantages of deep learning (neural network-based control algorithm) to solve the very challenges that has faced the ship berthing and unberthing sector over the past decades. Positive results from this project will ensure ships are safe from incurring damages during berthing and unberthing operations, lives and properties are protected and an overall reduced ship turnaround times are achieved at ports.

PROJECT SCOPE.

The project scope will entail trajectory tracking control system for berthing and unberthing mission and this will include the following components:

- Guidance system to generate desired trajectory for certain missions (berthing and unberthing)
- Control algorithm to make decision on the control signals sent to the vessel.

EXPECTED OUTCOME.

At the end of the project, it is expected that a numerical simulation is carried out to show how a PID and artificial neural network can be utilized to design a controller for trajectory tracking, berthing and unberthing mission. It is expected that the numerical simulation will cover

- i) MATLAB program development and simulation of a guidance system to generate and track a smooth reference trajectory path for a surface vessel. (Bluefin and Shioji Maru vessel).
- ii) PID and an NN-based controller for the trajectory tracking.

It is expected that the results of the simulation will suffice to show how artificial neural network controller and PID based controller can be used for automatic berthing and unberthing of a ship.

An experimental validation of the numerical simulation will be carried out and it is expected that the results of the experiment will validate the numerical simulations results. The ship should follow the reference trajectory which has been issued by the guidance system under the control of the ANN-controller and PID-based controller.

PROJECT WORK CHALLENGES.

Major challenges faced in this project includes

- i) Program errors and debugging of error.
- ii) Time management challenges
- iii) Unfinalized hydrodynamic coefficients of the mathematical model of Bluefin required for the simulation. This makes the Bluefin mathematical model unsuitable for simulation studies

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2.0 LITERATURE REVIEW

Prior works suggest a resolute global response to the challenges of trajectory tracking, berthing and unberthing of surface vessels. Several conventional feedback and linearized controls applied in designing controllers for trajectory tracking, berthing and unberthing missions of vessels flawed and were not robust. In recent times, intelligent control methods are fast becoming favorites with the artificial neural network being researched and developed further.

Yamato et al. (1990) was the first to develop a ship berthing controller using artificial neural network. Fujii and Ura (1991) studied the effectiveness of ANN-based controller in autonomous underwater vehicles using both the supervised and non-supervised learning. Yamato et al (1992) proposed an improvement to the ANN-based ship controllers where the inputs to the ANN-based controllers were the ship states, the desired states, and the control signal from past steps. This controller was considered multivariable and could be trained online. Simensen et al (1996) found that the use of artificial neural network controllers in ship control systems offered the advantages of an adaptive system in terms of the ability of the controller to learn and to handle variations in ship dynamics. They noted that in ship steering control, selected optimal settings of the conventional controller (like the PID-based controllers) vary when there are changes in operating conditions and dynamics of a ship. When there are significant variations in the ship's forward speed, water depth or the cargo weight, Conventional controllers lack an adaptive system behavior to match these possible variations in ship dynamics and as such do not suffice as an efficient ship control method. Like Yamato (1992), Zhang et al (1997) developed an online-trained backpropagation-based neural network controller to control ship berthing automatically. The controller could adjust its original parameters online by constant evaluation of its performance accuracy and was independent of the ship's mathematical model. This eliminated the need for offline training. They took considerations of wind disturbances and shallow water effects and demonstrated how robust the controller performed. Im and Hasegawa (2001) proposed a neural network with a parallel structure in the hidden layer which produced better results than the conventional ANN controller with a single hidden layer. Im and Hasegawa (2007) and Im (2009) applied a different approach that made it possible that the ship can start berthing from any point around the berthing region. This was considered as a selective ANN controller. Nguyen et al. (2007) proposed a dual ANN controller which used an adaptive interaction learning technique and the concept of a predetermined waypoints and a predetermined speed to control heading and speed during berthing mission. Tran and Im (2012) took ship maneuvering auxiliary devices; the side thrusters and tugboat into considerations and developed ANN controllers with a set of four outputs such that the rudder and propeller outputs guided the ship to the first target point and the bow thruster and stern tugboat guided the boat to the final destination. Ahmed and Hasegawa (2012) developed an artificial neural network control for automatic ship berthing. The used backpropagation technique where they trained two separate multi-layer feed-forward networks to control the ship rudder and propeller revolution. They validated their simulation results experimentally using the Esso Osaka 3-m model. Ahmed and Hasegawa (2013) suggested the use

of virtual windows together with non-linear programming methods as a way of creating consistent teaching data for training the network in an ANN. The PD component of a PID controller was implemented together with this to keep the ship on some sought of imaginary reference line after any change in heading caused by the ANN controller. This was particularly suitable for gusty winds. This ANN controller concept was found valid for ship berthing and there were no recorded cases of disturbances. An adaptive backstepping controller has been suggested by Park and Kim (2014). This can be used for pulling or pushing a cruise ship which is subject to wind and can be applied only in limited areas and only when the ship is near the desired berthing point (Berthing pier) and the lateral forces can be controlled to berth the ship in a crabwise motion.

Several other works on trajectory control and automatic berthing and unberthing of ships saw researchers use the fuzzy logic and its hybrid. Morrison (2015) investigated the feasibility of developing an autonomous ship control system using fuzzy logic-based controllers and low-cost sensors on a model-scale of the Bluefin vessel. The control algorithm did not suffice for the proposed project and challenges relating to the hardware also hindered the success of the work. Pathan, Unar and Memon (2016) developed a fuzzy logic controller for an autopilot of an oil tanker. Non-linear mathematical model of an oil tanker was used with the parameters varying with depth. The triangular membership function was used for fuzzification and the centroid method for defuzzification. The controller's performance was found satisfactory for trajectory control and course changing missions both in shallow and deep waters. Liang (2018) developed a trajectory control using both the fuzzy logic theorem and the PID controller. Simulations was carried out using the non-linear model of Shioji Maru and on a proposed mathematical model of Bluefin model. Simulations results showed that the PID-based control algorithm outperformed the fuzzy logic-based algorithm.

The scale model of Bluefin is the prototype of the AMC research vessel which has been used at the Australian Maritime College for research in the areas of ship control and maneuvering. Recent works have seen researchers like Jiang (2016), who modelled and simulated the hydrodynamic characteristics of the bluefin model scale using low cost sensors. An estimate of the hydrodynamic coefficients of the model scale were made using the least square estimation algorithm and the recursive least square estimation algorithm. Hao's (2017) research led to an improvement on the mathematical model and the hydrodynamic coefficients of the proposed mathematical model for the Bluefin model scale. Li (2018) developed PID based trajectory tracking system using the concept of waypoint guidance system for Bluefin model. He investigated different scenarios including; PID control with 2 controls (CPP and rudder), with 3 controls (CPP, rudder and bow thruster), 3 other controls (rudder, stern thruster and bow thruster) and 4 controls (CPP, stern thruster, bow thruster and Rudder). From his findings, he concluded that using 3 controls (rudder, stern thruster and bow thruster), offered the desired results but since the scale model of bluefin is not equipped with a stern thruster, the PID based trajectory control which utilized only rudder and

speed control sufficed for the bluefin model. Haoliang (2019) made the first successful attempt at developing and conducting experiments for a Neural network-based trajectory control of the model scale of Horn at the AMC. However, the experimental results showed large cross track errors through the trajectory path which suggested the need for improvements on the algorithm.

3.0 THE PROJECT DEVELOPMENT

3.1 DESCRIPTION OF PLATFORM BOAT AND MATHEMATICAL MODELS FOR DESIGN

The Bluefin is the research and training vessel at the Australian Maritime college and hence, the scale model of Bluefin has been initially proposed for this research. Due to the global outbreak of the COVID-19, fellow researchers are unable to finalize the hydrodynamic coefficients of the mathematical model of Bluefin and this has hindered the author from conducting this research using the Bluefin's mathematical model. The finalized mathematical model of Shioji Maru vessel (Nguyen 2014) has been selected as an alternative. Table 1 highlights the principal particulars of the Shioji Maru and the Bluefin vessel.

Table 1. Description of model scaled boat Bluefin

	Bluefin (Scale Model)	Bluefin (Full scale)	Shioji Maru (Full scale)
Length overall	1.725m	35m	
LBP	1.569m	32m	46m
Breadth	0.5m	-	10m
Mass	64.55kg	-	$6.7418 \times 10^4 \text{ kgs}^2/\text{m}$
Draft	0.196m	4.4m	2.85m
Propeller	Fixed pitch propeller (rpm)	cpp	cpp
Thruster	Bow thruster	Bow thruster	Bow and Stern thruster

Ship motion in all 6 degrees of freedom is a translational motion in three directions; surge, sway and heave and a rotational motion in three axes; roll, yaw and pitch. However, Motions in pitch and heave can generally be neglected for conventional surface ships and as a result, four degrees of freedom will be considered for the surface ship model.

In a body-fixed frame, using the standard notation given in SNAME (1950), the forces, moments and the linear-angular velocity vector are shown in Figure 1 and described as follows.

$$\dot{\delta} = \frac{\delta_c - \delta}{|\delta_c - \delta| T_{Rud} + a} \quad (13)$$

$$\dot{\theta} = \frac{\theta_c - \theta}{|\theta_c - \theta| T_{cpp} + a} \quad (14)$$

Further details on the mathematical model of Shioji Maru vessel can be found in (Nguyen 2014).

3.2 INVESTIGATION OF THE OPEN LOOP CHARACTERISTICS OF THE SHIOJI MARU SHIP MODEL

To investigate the maneuverability and the response of the ship model, the open loop characteristics of the Model-scale Bluefin and Shioji Maru has been investigated. These simulation studies evaluate the response of the ship to the changes in the ship's actuators. These includes, change in the propeller's rpm/CPP, change in the rudder angle and the bow thruster notch. The response will verify if the ship model will suffice for the trajectory tracking simulation.

SIMULATION RESULTS

Table 2. Test condition: 1 (Figure 2)

Rudder angle	0degrees
Propeller pitch (Cpp)	15 degrees
BT	0Kg

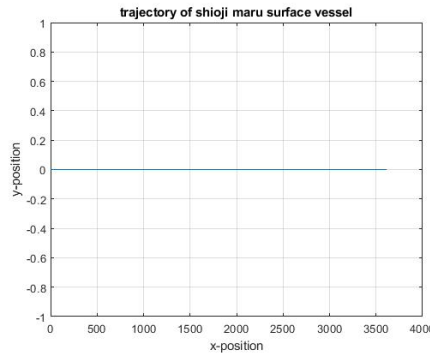


Figure 26. Maneuvering test results, 0 degrees rudder angle, 15deg cpp, BT 0kg.

Figure 2 shows the simulation results of the proposed model under open loop conditions. For test condition 1 shown on Table 2, at a rudder angle of zero, a propeller pitch of 15cpp and 0kg on the bow thruster, a straight-line trajectory is obtained. This result is expected and shows that the vessel can maintain a straight-line trajectory in the absence of disturbances.

Table 3. Test condition: 2 (Figure 3)

Rudder angle	20degrees
Propeller pitch (Cpp)	15 degrees

BT	0Kg
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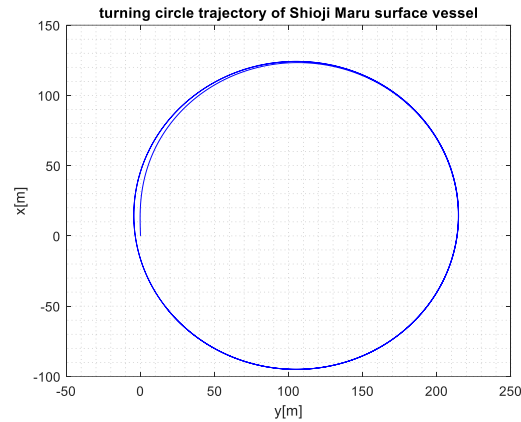


Figure 27. Maneuvering test results, 20 degrees rudder angle, 15deg cpp, BT 0kg.

Table 4. Test condition: 3 (Figure 4)

Rudder angle	-20degrees
Propeller pitch (Cpp)	15 degrees
BT	0Kg

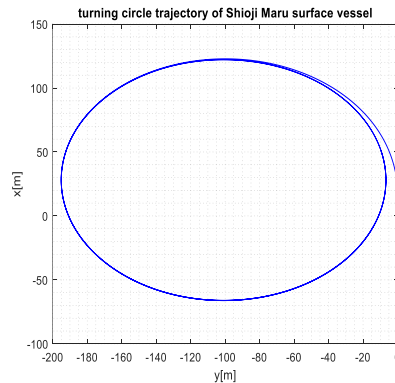


Figure 28. Maneuvering test results, -20 degrees rudder angle, 15deg cpp, BT 0kg.

For test condition 2 and 3 shown on Table 3 and 4 respectively, Figure 3 and 4 shows a steady turning circle produced when the rudder is deflected to 20 degrees and -20 degrees respectively. The thrusters are not used, and the propeller is run at a constant pitch of 15°. The result shows a steady and a smooth turning circle radius. This confirms the ability of the ship to change and maintain heading when the rudder angle is deflected to the starboard or port-side.

Table 8. Test condition: 4 (Figure 5)

Rudder angle	-20degrees
--------------	-------------------

Propeller (Cpp)	pitch	15 degrees
BT		25Kg

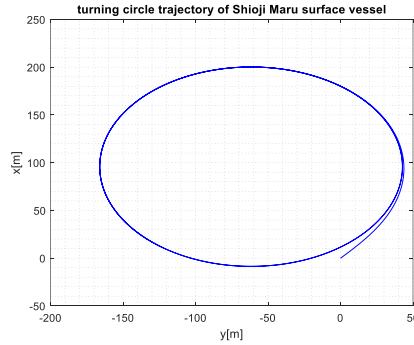


Figure 29. Maneuvering test results, 20 degrees rudder angle, 15deg cpp, BT 25kg

Table 9. Test condition: 5 (Figure 6)

Rudder angle		-20degrees
Propeller (Cpp)	pitch	15 degrees
BT/ST		25Kg/-25Kg

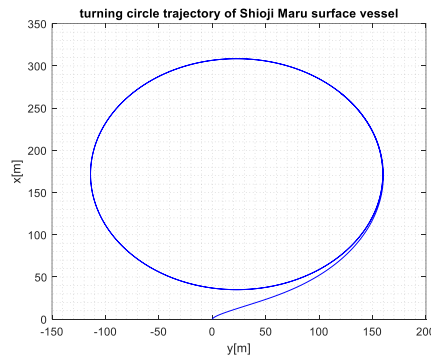


Figure 30. Maneuvering test results, 20 degrees rudder angle, 15deg cpp, BT -25kg.

For test condition 5 shown on Table 6, Figure 6 shows the effect of the bow thruster in the ship's maneuvering. At a rudder angle of -20 degrees, a bow thruster notch of -25kg and a propeller pitch of 15°, the ship trajectory shows a turning circle with a relatively smaller radius compared to the actuation by rudder and propeller alone. This results to an unsmooth trajectory due to the added turning moments created by the bow thruster. figure 6 shows a smaller turning circle due to the added effect of the stern thruster.

3.3 CONTROL ALGORITHM

3.3.1 Structure of the control system

The proposed ship control algorithm has been developed in modules to reflect the different functional units of the system which are required to work in synergy to accomplish the control objective. Figure 7 shows the functional block diagram of the proposed control system algorithm.

The concept suggested by Nguyen (2007) is adopted such that the guidance system generates the reference trajectory using a predefined waypoint information and the controller cascades make the control decisions using the reference trajectory information as the setpoints and the feedback(ship states) from the ship navigation system as the process variable. The controller cascades consist of the NN-based autopilot, the speed controller, and the position controller. A description of the algorithm for the separate modules of the control system is presented in subsequent sections.

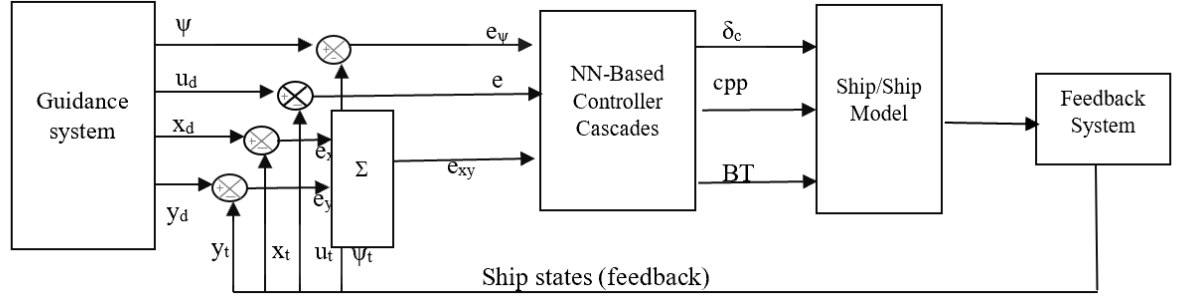


Figure 31. functional block diagram of the control system for trajectory tracking, berthing and unberthing mission

3.3.2 The way-point guidance system

The way-point guidance system using the line-of-sight algorithm for path control as proposed by Fossen (2002) is adopted by the author. A set of predefined waypoint information is used to generate the reference trajectory for the ship. If there are 'k' number of waypoints along the reference path, the waypoint information consists of the earth-fixed x and y-coordinates (positions) and the surge velocities and can be expressed as:

$$\{(y_1, x_1, U_1), (y_2, x_2, U_2), \dots, (y_k, x_k, U_k), \} \quad (15)$$

The guidance system task will primarily consist of heading generation using the line-of-sight algorithm, generation of the reference trajectory using the 1-DOF T-K model and guidance trajectory tracking using the basic PID control law. The functional block diagram of the guidance system is as shown in Figure 8.

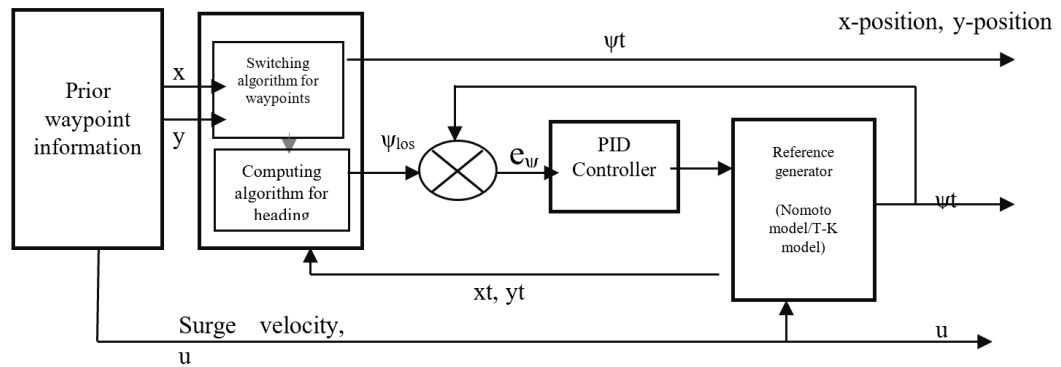


Figure 32 functional block diagram of the guidance system

A functional description of the guidance system modules explains how the algorithm computes and generates the reference trajectory from the waypoint information.

3.3.3 The Line of sight algorithm

To achieve the control objective for the trajectory tracking, a time varying line of sight algorithm suggested by Fossen (2002) is used to compute the desired heading. Considering Figure 9, if the control objective of the guidance system requires that the reference trajectory is defined from waypoint (y_{k-1}, x_{k-1}) to waypoint (y_k, x_k) . At time, 't', when the reference vector is at 'P', the line-of-sight vector is the vector that points from the body-fixed origin, (y_t, x_t) to the next waypoint, (y_k, x_k) . Thus, the desired heading, $\psi_{los}(t)$ at time (t), to the guidance system course autopilot can be expressed as,

$$\psi_{los}(t) = \text{atan2}(y_k - y(t), x_k - x(t)) \quad (16)$$

The four-quadrant inverse tangent function $\text{atan2}(\Delta y, \Delta x)$ defines the limits of the desired heading angle as:

$$-\pi \leq \psi_{los}(t) \leq \pi \quad (17)$$

Thus, the computed desired head $\psi_{los}(t)$ is taken as the setpoint to the guidance system course autopilot.

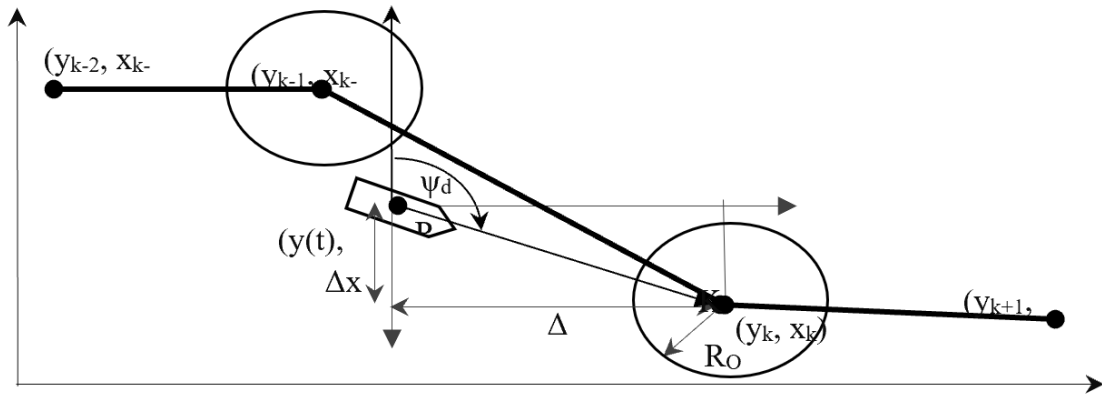


Figure 33. Illustration of the line-of-sight algorithm

3.3.4 The Guidance system course autopilot.

The guidance system course autopilot is designed using the PID control law and is of the form,

$$\delta_c = K_p(\psi_{los} - \psi_t) + K_i \int_0^t (\psi_{los} - \psi_t) + K_d \frac{d(\psi_{los} - \psi_t)}{dt} \quad (18)$$

ψ_{los} is the desired heading computed using the LOS algorithm, ψ_t is the actual heading of the guidance system reference generator, K_p , K_i , and K_d are the PID gain parameters of the guidance system course autopilot and δ_c is the outputted commanded rudder from the guidance system autopilot.

For the reference trajectory under study, the PID gain parameters are selected as $K_p=59.5$, $K_i=0.5$, $K_d=0.09$.

3.3.5 Generation of the reference trajectory using the first order Nomoto model

To generate the desired reference trajectory, the course dynamics is modelled as:

$$\dot{\psi}_d = r_d \quad (19)$$

$$T\dot{r}_d + r_d = K\delta_c \quad (20)$$

where T and K are ship's manoeuvrability indices selected as, $K = 0.11[s^{-1}]$, $T = 7.5[s]$ for medium and low speed range of the Shioji Maru vessel. (Nguyen 2014).

The commanded rudder angle δ_c is selected from output of the guidance system course autopilot defined in equation (18):

$$\delta_c = K_p(\psi_{los} - \psi_t) + K_i \int_0^t (\psi_{los} - \psi_t) + K_d \frac{d(\psi_{los} - \psi_t)}{dt} \quad (21)$$

The desired reference trajectories of the ship can be modelled as:

$$\dot{x}_d = U_d \cos \psi_d \quad (22)$$

$$\dot{y}_d = U_d \sin \psi_d \quad (23)$$

where $U_d > 0$ is the desired surge velocity.

Numerical integration of the ODEs in equation (19) through (23) will yield the desired smooth reference trajectory $(y_d(t), x_d(t), \psi_d(t))$ with speed assignment $U_d(t)$ (Fossen 2002). To generate the reference trajectory, a change of heading is required at every new waypoint. A switching algorithm to achieve this is described in the next section.

3.3.6 The waypoint Switching algorithm

The switching algorithm is developed to allow the guidance system reference vector to change heading by switching to a new waypoint when certain minimum tracking conditions have been met. Consider Figure 3. At point K , to select new waypoints, (y_k, x_k) , the reference vector must be within the circle of acceptance, R . Hence the switching algorithm is selected as:

$$\text{if } \sqrt{((y_k - y_t)^2 + ((x_k - x_t)^2} \leq R \quad (24)$$

$$\text{waypoint}_{number} = \text{waypoint}_{number} + 1 \quad (25)$$

$$R = 2L_{pp}. \quad (26)$$

where L_{pp} is the ship length (Shioji Maru's length=46m)

3.3.6 The basic PID-based trajectory tracking algorithm

According to Fossen (2002), the PID control law is used to design basic set of controllers to compare and evaluate the performance of the proposed Neural network-based controllers. The PID-based course autopilot is of the form:

$$\delta_c = K_p(\psi_d - \psi_t) + K_i \int_0^t (\psi_d - \psi_t) + K_d \frac{d(\psi_d - \psi_t)}{dt} \quad (27)$$

the angle between the desired heading from the guidance system ψ_d and the time-varying ship heading ψ_t is the input to PID-based course autopilot, K_p , K_i , and K_d are carefully selected PID gain parameters and δ_c , the controller output is selected as the commanded rudder angle of the ship. Similarly, the PID-based speed controller is of the form

$$cpp_c = K_{ps}(U_d - U_t) + K_{is} \int_0^t (U_d - U_t) + K_{ds} \frac{d(U_d - U_t)}{dt} \quad (28)$$

where K_{ps} , K_{is} , and K_{ds} are carefully selected PID gain parameters for the speed controller, U_d is the desired surge velocity, U_t is the time varying surge velocity of the ship.

3.3.7 The artificial neural network concept in control systems.

The artificial neural network as a mathematical model that seeks to simulate the structure and functionalities of biological neural networks. (Andrej et al 2011). The neural network algorithm works by utilizing optimized weight and bias values which are attained by a learning process to approximate outputs when inputs values are presented. For every control cycle, when the desired heading, position and surge velocity of a ship are presented to the network, the NN algorithm should accurately adjust the ship's commanded rudder angle, bow thruster and the propeller rpm to reduce the error between the desired states (heading, position and surge velocity) and the current ship states to keep the ship within the desired reference trajectory. Unar (1999) expressed the NN algorithm as:

$$y = f\left(\sum_{i=1}^p (w_i x_i + b)\right) \quad (29)$$

where f is the activation function, p is the number of inputs, x_i is the NN input, b is the bias and w_i is the weight.

The proposed algorithm for the NN-based controllers is the multilayer perceptron with the backpropagation learning technique. For each controller, the network inputs are selected as the control error and the error derivative. In the forward pass, the approximation requires that the network inputs are multiplied by the weight values and then shifted by the bias value. The carefully selected activation function squashes the output within the function limits to give the neuron output.

In the learning process, the backpropagation algorithm backpropagates the errors by computing the cost function from the NN prediction error and computing the error sensitivities. During backpropagation, a carefully selected learning rate parameter determines the rate at which these errors are backpropagated to update the network weights and the biases. The updated weights and bias are used in the next control cycle to make new predictions such that the prediction error becomes minimized. This is repeated at every control cycle.

3.3.8 Algorithm for developing and training the proposed ANN-based controller.

The flow chart 1 below presents the control algorithm for the NN-based controllers. The weights, bias and learning rate are initialized and at every control cycle, the ANN-forward pass is conducted using the weights and the bias to calculate the outputs of neurons in the hidden layer and at the output layer. To efficiently achieve the control objective, the cost function and the error sensitivities are computed to backpropagate the errors. The learning rate and the error sensitivities are used to update the weights and the biases which are used subsequently in the next control cycle to minimize the cost function. The control cycle is repeated through the trajectory tracking, automatic berthing and unberthing process.

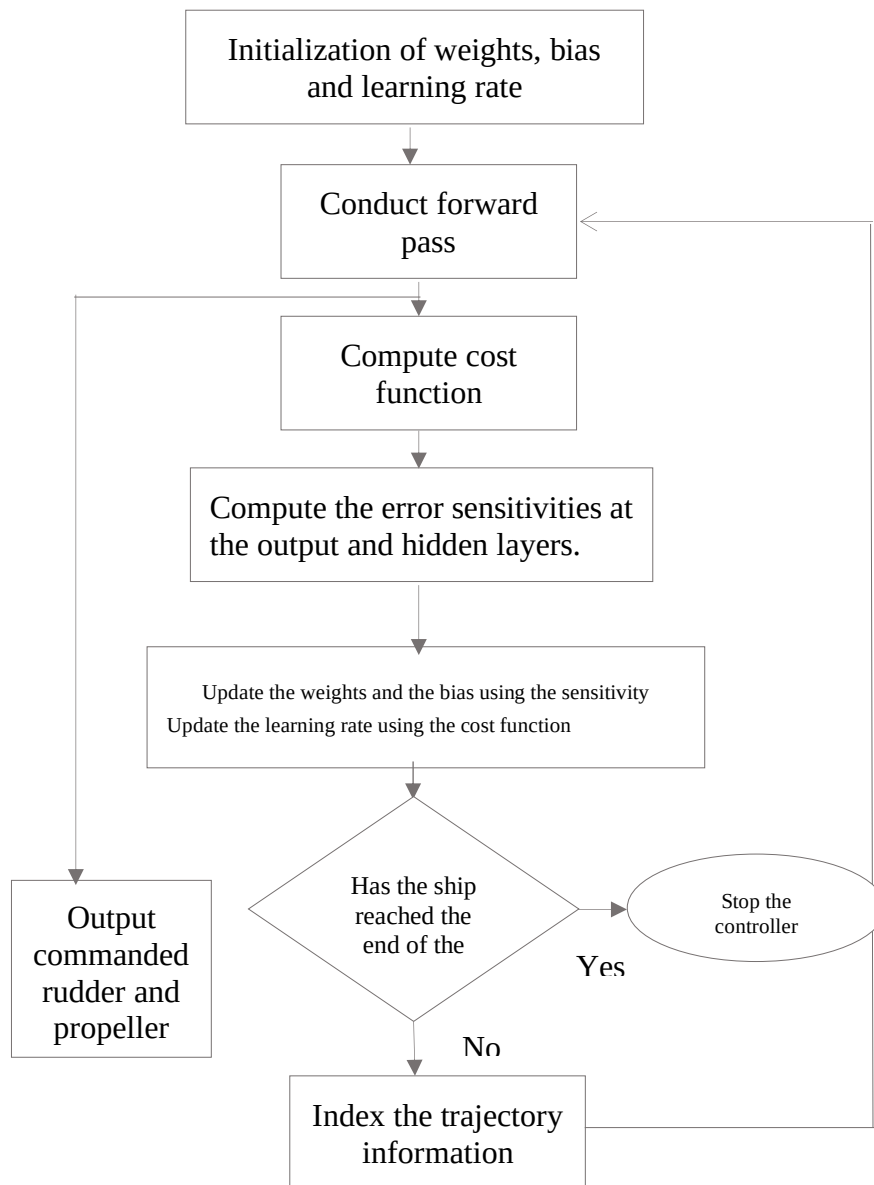


Chart 1. Proposed NN algorithm.

3.3.9 The proposed neural network architecture and algorithm for autopilot, speed controller and position controllers.

The neural network-based control algorithms have been in use over the years. The architecture as suggested by Vink (2017) is adopted in designing the NN-based controllers. The proposed architecture for the NN-based autopilot and speed controllers is shown in Figure 10.

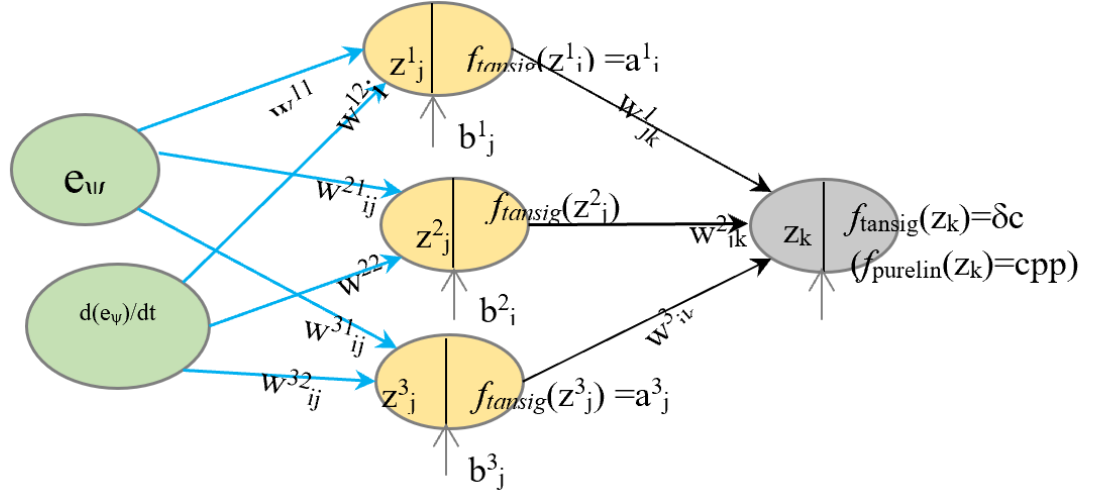


Figure 34. Proposed neural network architecture.

The proposed architecture of the NN-based controllers is the multilayer perceptron neural network. A three-layer feedforward network with 3 neurons at each layer (to improve the controller performance) is proposed. At the input layer of the NN-based autopilot, the input vectors are selected as the ship heading error and the derivative of the heading error.

$$e_\psi = \psi_d - \psi_t \quad (30)$$

$$\dot{e}_\psi = \frac{d(\psi_d - \psi_t)}{dt} \quad (31)$$

$$x = [e_\psi \ \dot{e}_\psi] \quad (32)$$

Similarly, for the speed controller, the input vectors are the surge velocity error (error between the desired surge velocity and the actual surge velocity of the ship) and derivative of the surge velocity error.

$$e_u = u_d - u_t \quad (33)$$

$$\dot{e}_u = \frac{d(u_d - u_t)}{dt} \quad (34)$$

$$x = [e_u \ \dot{e}_u] \quad (35)$$

Considering the forward pass to from the input layer to the hidden layer, the outputs for all three neurons before and after activation are expressed respectively as:

$$z_j = \sum(w_{ij}x) + b_j \quad (35)$$

$$a_j = f_{zi}(z_j) = f_{zi}(\sum(w_{ij}x) + b_j) \quad (36)$$

where, f_{zi} is a non-linear activation function, tan-sigmoid defined as:

$$f(z) = \frac{2}{1 + e^{-2z}} - 1 \quad (37)$$

The tan sigmoid activation function limits the outputs of the hidden-layer neurons from -1 to 1. The tan sigmoid activation function has been selected because it learns faster in a multilayer perceptron network than the logistic function (Haykin 1994).

In the second stage of the forward pass, the activated neuron-outputs of the hidden layer a_j becomes the inputs for the next forward pass. From the hidden layer to the output layer, the neuron-outputs before and after activation are expressed respectively as:

$$z_k = w_{jk}a_j + b_k \quad (38)$$

$$y = f_{zj}(z_k) = f_{zj}(\sum(w_{jk}a_j) + b_k) \quad (39)$$

In the NN-based autopilot, the activation function in the second stage of the forward pass is selected as the tan sigmoid activation function and the output 'y' is selected as the commanded rudder angle of the ship.

$$\delta_c = y = f_{zj_tansigmoid}(z_k) \quad (40)$$

In the NN-based speed controller, the activation function in the second stage of the forward pass is selected as the purelin activation function and the output 'y' from equation (39) becomes the propeller cpp for the ship.

$$cpp_c = y = f_{zj_purelin}(z_k) \quad (41)$$

While the hyperbolic-tangent activation function is employed at the output layer of the linear activation function, 'purelin' is used at the output layer of the speed controller. The tan sigmoid function has a maximum value of 1 and does not suffice to approximate the continuous values of the propeller cpp/rpm. The 'purelin' activation function produces values above 1 to control the cpp and to map the ship's surge velocity to the desired surge velocity.

For i, j, k denoting input, hidden and output layer respectively, for each of the NN-based autopilot and speed controllers, the forward pass is summarised as:

$$z_j^1 = (e_\psi \cdot w_{ij}^{11} + \dot{e}_\psi \cdot w_{ij}^{12}) + b_j^1 \quad (42)$$

$$z_j^2 = (e_\psi \cdot w_{ij}^{21} + \dot{e}_\psi \cdot w_{ij}^{22}) + b_j^2 \quad (43)$$

$$z_j^3 = (e_\psi \cdot w_{ij}^{31} + \dot{e}_\psi \cdot w_{ij}^{32}) + b_j^3 \quad (44)$$

$$a_j^1 = f_{tansig}(z_j^1) \quad (45)$$

$$a_j^2 = f_{tansig}(z_j^2) \quad (46)$$

$$a_j^3 = f_{tansig}(z_j^3) \quad (47)$$

$$z_k = (a_j^1 \cdot w_{jk}^1 + a_j^2 \cdot w_{jk}^2 + a_j^3 \cdot w_{jk}^3) + b_k \quad (48)$$

$$\delta_c = f_{tansig}(z_k) \quad (49)$$

$$cpp = f_{purelin}(z_k) \quad (50)$$

The algorithm of the feedforward network of the NN can only make prediction of the setpoints, however, to minimize the tracking error constituted by the predictions, the backpropagation learning algorithm is proposed.

3.3.10 Minimization of the tracking error by back-propagation learning algorithm

The proposed ANN-based controller is trained by the back-propagation rule. The back-propagation algorithm seeks to adjust the weights and biases of the ANN-network to minimize the sum of the squared errors of the network (cost function). This is achieved by continuous adjustment of the values of the network weights and biases in the direction of steepest descent with respect to error. The objective of the algorithm is to compute the partial derivatives of the cost/loss function with respect to weights and bias in the network. Hence, the loss function gives a measure of the ANN-based controller prediction error.

Considering the parameters defined as,

i =input

j =hidden layer

k =output layer

x =input vectors

w_{ij} =weights from neuron i to neuron j

b =bias value of neuron

z =neuron output before activation

$f(z)$ =activation function of neuron

a =desired output value of neuron after activation

y =prediction of the NN-controller at the network's output layer (commanded rudder angle, propeller cpp)

ψ_t =actual heading of the ship based on the NN-based autopilot control decision

u_t = actual surge velocity of the ship based on the NN-speed controller decision.

η =learning rate

The backpropagation is carried out from the output layer backwards towards the input layer.

Considering the NN-based autopilot, if the heading-prediction made by the NN output layer on the non-linear mathematical model of the ship is ψ_t and the desired heading from the guidance system is ψ_d , then the cost function becomes:

$$J_\psi = \left(\frac{1}{2}\right) (\psi_d - \psi_t)^2 = \left(\frac{1}{2}\right) (e_\psi)^2 \quad (51)$$

Similarly, for the speed controller, the cost function is:

$$J_u = \left(\frac{1}{2}\right) (u_d - u_t)^2 = \left(\frac{1}{2}\right) (e_u)^2 \quad (52)$$

Given that the control objective is to minimize the loss function to ensure that the ship tracks the reference trajectory perfectly, taking a partial derivative of the loss function (backpropagation) in each of the neural network layers with respect to the weights, $\frac{\partial J}{\partial w_i}$ will give a gradient of the curve in the direction of the required weight and bias adjustment.

If i, j, k denotes input, hidden and output layers respectively, then in backpropagating the errors from the output layer to the hidden layer, the partial derivative of the loss function J with respect to the weights in the hidden layer is

$$\frac{\partial J}{\partial w_{(jk)}} = \frac{\partial J}{\partial y} \cdot \frac{\partial y}{\partial z_k} \cdot \frac{\partial z_k}{\partial w_{(jk)}} \quad (53)$$

and the partial derivative of the loss function J with respect to the output layer bias is,

$$\frac{\partial J}{\partial b_{(k)}} = \frac{\partial J}{\partial y} \cdot \frac{\partial y}{\partial z_k} \cdot \frac{\partial z_k}{\partial b_{(k)}} \quad (54)$$

From equation (51), (39) and (38) respectively,

$$\frac{\partial J}{\partial y} = e_\psi \quad (55)$$

$$\frac{\partial y}{\partial z_k} = (1 - y^2) \quad \left(\frac{\partial y}{\partial z_k} = 1, \text{ for the speed controller with purelin activation function} \right) \quad (56)$$

$$\frac{\partial z_k}{\partial w_{(jk)}} = a_j \quad (57)$$

$$\frac{\partial z_k}{\partial b_{(k)}} = 1 \quad (58)$$

Substituting equation (55), (56) and (57) into equation (53) gives

$$\frac{\partial J}{\partial w_{(jk)}} = e_\psi \cdot (1 - y^2) \cdot a_j = S_{kj} \times a_j \quad (59)$$

Similarly, equation (54) becomes

$$\frac{\partial J}{\partial b_{(k)}} = e_\psi \cdot (1 - y^2) \cdot 1 = S_{kj} \cdot 1 \quad (60)$$

where S_{kj} , describes the backpropagation sensitivity from the output layer to the hidden layer.

If the learning rate is defined as 'eta', the weight, and bias updates in the 3 neurons between the output layer and the hidden layer are

$$w_{jk}^1 = w_{jk}^1 + S_{kj} \times a_j^1 \times \text{eta} \quad (61)$$

$$w_{jk}^2 = w_{jk}^2 + S_{kj} \times a_j^2 \times \text{eta} \quad (62)$$

$$w_{jk}^3 = w_{jk}^3 + S_{kj} \times a_j^3 \times \text{eta} \quad (63)$$

$$b_k = b_k + S_{kj} \times \text{eta} \quad (64)$$

In backpropagating the errors from the hidden layer to the input layer, the partial derivative of the loss function with respect to the weights between the hidden layer and the input is,

$$\frac{\partial J}{\partial w_{(ij)}} = S_{kj} \cdot \frac{\partial z_{(k)}}{\partial a_{(j)}} \cdot \frac{\partial a_{(j)}}{\partial z_{(j)}} \cdot \frac{\partial z_{(j)}}{\partial w_{(ij)}} \quad (65)$$

from equation (33), (31) and (30) respectively

$$\frac{\partial z_{(k)}}{\partial a_{(j)}} = w_{jk} \quad (66)$$

$$\frac{\partial a_{(j)}}{\partial z_{(j)}} = (1 - a_j^2) \quad (67)$$

$$\frac{\partial z_{(j)}}{\partial w_{(ij)}} = x \quad (x = \text{NN inputs}) \quad (68)$$

Substituting equation (66), (67) and (68) into equation (65) yields the weight updates, given as

$$\frac{\partial J}{\partial w_{(ij)}} = S_{kj} \cdot w_{jk} \cdot (1 - a_j^2) \cdot x \quad (69)$$

$$\frac{\partial J}{\partial w_{(ij)}} = S_{ji} \cdot x \quad (70)$$

where $S_{ji} = S_{kj} \cdot w_{jk} \cdot (1 - (a_j)^2)$, is the sensitivity between the hidden layer and the input. Similarly, the bias update is obtained as,

$$\frac{\partial J}{\partial b_{(j)}} = S_{ji} \quad (71)$$

for all 6 neurons between the hidden layer and the input layer, if the learning rate is selected as 'eta', the weight and bias updates are

$$w_{ij}^{11} = w_{ij}^{11} + S_{ji}^1 \times x(1) \times \text{eta} \quad (72)$$

$$w_{ij}^{12} = w_{ij}^{12} + S_{ji}^1 \times x(2) \times \text{eta} \quad (73)$$

$$w_{ij}^{21} = w_{ij}^{21} + S_{ji}^2 \times x(1) \times \text{eta} \quad (74)$$

$$w_{ij}^{22} = w_{ij}^{22} + S_{ji}^2 \times x(2) \times \text{eta} \quad (75)$$

$$w_{ij}^{31} = w_{ij}^{31} + S_{ji}^3 \times x(1) \times \text{eta} \quad (76)$$

$$w_{ij}^{32} = w_{ij}^{32} + S_{ji}^3 \times x(2) \times \text{eta} \quad (77)$$

$$b_j^1 = b_j^1 + S_{ji}^1 \times \text{eta} \quad (78)$$

$$b_j^2 = b_j^2 + S_{ji}^2 \times \text{eta} \quad (79)$$

$$b_j^3 = b_j^3 + S_{ji}^3 \times \text{eta} \quad (80)$$

where, eta = learning rate of the back-propagation algorithm.

The backpropagation process through the NN-based autopilot network is shown in Figure 11 below

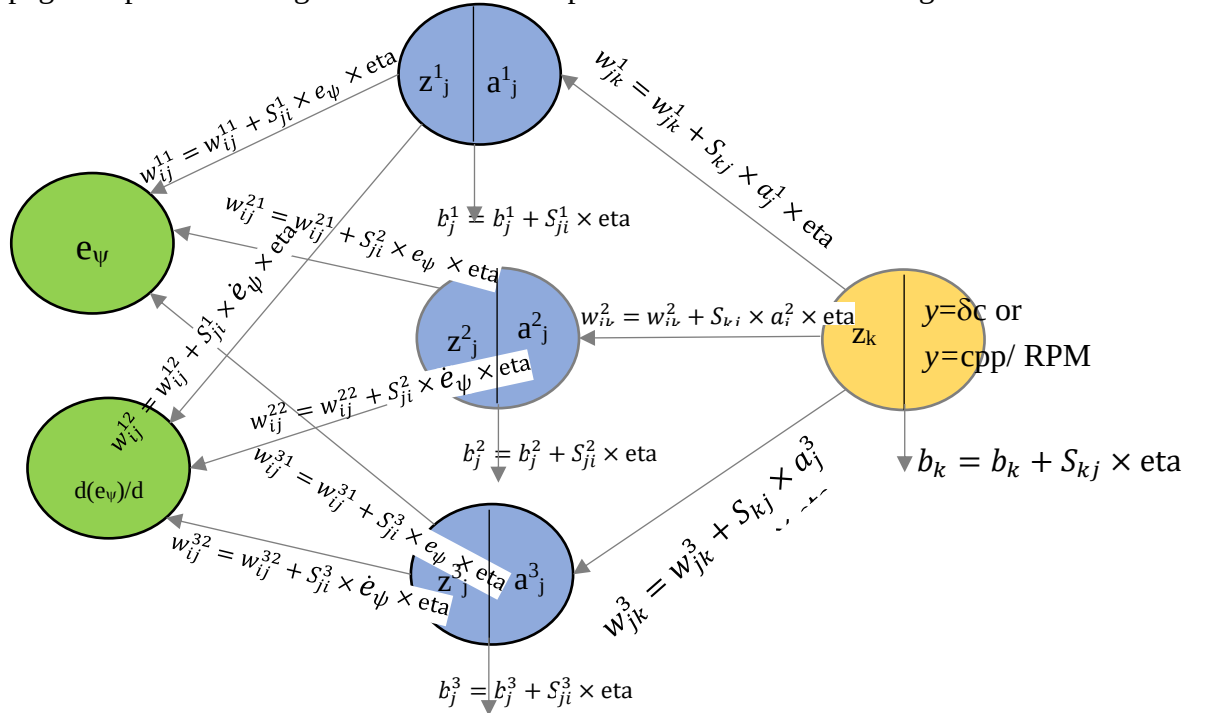


Figure 35. Backpropagation through the NN-based autopilot network

3.3.11 Proposed learning rate algorithm for the neural network-based heading controller.

Evaluation of the NN-based controller shows that in tracking lower heading angles, higher learning rate values are required while higher heading angles require lower values of learning rate. This is illustrated on Table 7. Through iterative process, a learning rate database is developed such that the predefined learning rate suffices to track the desired heading angle at every control cycle. The learning rate algorithm is incorporated into the guidance system such that the guidance system generates the reference heading (trajectory) alongside the predetermined learning rate required to track the desired path for every control cycle.

Table 10. Illustration of variation of learning rate with heading angle

Heading angle	Learning rate, eta
5°	0.0005
10°	0.0002
15°	0.0001
20°	0.0001
25°	0.0001
30°	0.000048
35°	0.000046
40°	0.000043

The learning rate algorithm is given thus,

$$\text{if } \psi_d < 5(\pi/180)$$

$$\eta = \eta_n$$

$$\text{if } \psi_d \geq 10(\pi/180)$$

$$\eta = \eta_{1+n}$$

•

•

•

$$\text{if } \psi_d \geq 360(\pi/180)$$

where $\eta_n, \eta_{1+n}, \eta_{2+n}$ are pre – determined learning rates

ψ_d = desired heading

3.3.12 Position control using the bow thruster

The earth-fixed position errors for the trajectory control is given by

$$x_e = x - x_t \quad (81)$$

$$y_e = y - y_t \quad (82)$$

As suggested by Lekkas (2014), the position error for the vessel position (x, y) in vessel-fixed frame is given by,

$$\begin{bmatrix} x_e \\ y_e \end{bmatrix} = R^T(psi) \begin{bmatrix} x - x_t \\ y - y_t \end{bmatrix} \quad (83)$$

where R^T is a transformation matrix given as

$$R^T = \begin{bmatrix} \cos(psi) & -\sin(psi) \\ \sin(psi) & \cos(psi) \end{bmatrix} \quad (84)$$

Hence, the deviation in the transverse direction or the cross-track error can be given as,

$$x_e = (x - x_t) \cos(psi) + (y - y_t) \sin(psi). \quad (85)$$

To minimize the position error, x_e is selected as an input to the bow thruster actuator.

3.3.13 Improvement of the back-Propagation Neural Network Learning Algorithms using adaptive gain values.

Nawi, Ransing and Hamid (2010) proposed an ANN algorithm with adaptive gain values for the back-propagation algorithm. This significantly reduces the training time and consequently improves the performance of the ANN.

In principle, an activation function limits the amplitude a neural network neuron output, usually in a predefined range in a closed unit interval of [0, 1] as in the logistic sigmoid activation function or [-1, +1] as in the hyperbolic tangent activation function.

From equation (37), the hyperbolic activation function given by,

$$f(z) = \frac{2}{1+e^{-2z_j}} - 1 \quad (88)$$

Nawi, Ransing and Hamid (2010) proposed that equation (88) can be re-written as,

$$f(z) = \frac{2}{1+e^{-2gz_j}} - 1 \quad (89)$$

where ‘g’ is introduced as the gain value for the proposed ANN.

Nawi, Ransing and Hamid (2010) demonstrated that numerical value of the gain parameter, ‘g’ directly affects the slope of the activation function used in the neural network.

For gain values greater than unity, ($g \gg 1$), the activation-function tends towards ‘step function’ whereas for smaller gain values ($0 < g \ll 1$), the output values of the activation function changes steadily over the closed unit interval of $[-1,1]$ (for the hyperbolic activation function) and approximates a linear function thereof.

While the gain value has significant effects on the training speed, it is proven that higher gain values causes instability in the network.

Nawi, Ransing and Hamid (2010) in their work suggested an expression to update the gain values for the output and hidden layers as a function of the output error loss-function.

Recall from equation (51), the loss function of the ANN-based heading controller is given as,

$$J_\psi = \left(\frac{1}{2}\right) (e_\psi)^2 \quad (90)$$

where the ship heading error,

$$e_\psi = \psi_d - \psi_t \quad (91)$$

If the gain parameters at the output layer and hidden layer is given by g_k and g_j respectively,

then the proposed updates to the gain parameters are given by,

$$\Delta g_k = \text{eta} \left(-\frac{dJ_\psi}{dg_k} \right) \quad (92)$$

$$\Delta g_j = \text{eta} \left(-\frac{dJ_\psi}{dg_j} \right) \quad (93)$$

Derivative of the loss function with respect to the gain parameters at the output and hidden layers at every control cycle gives an update to the proposed gain parameters, thus, at the output layer,

$$\frac{dJ_\psi}{dg_k} = -e_\psi \times z_k \times (1 - y^2) \quad (94)$$

at the hidden layer,

$$\frac{dJ_\psi}{dg_j} = -\sum (g_k w_{jk} (1 - y^2) e_\psi) \times (1 - a_j^2) \times z_j \quad (95)$$

recall, y is the output of the ANN and z_k is the output layer neuron output before activation, z_j is the hidden layer neuron output before activation and a_j is the hidden layer activated neuron output.

Thus, the proposed gain updates can be summarized as,

$$g_k = g_k + \text{eta} (e_\psi \times z_k \times (1 - y^2)) : \quad \text{at the output layer} \quad (96)$$

$$g_j = g_j + \text{eta} (\sum (g_k w_{jk} (1 - y^2) e_\psi) \times (1 - a_j^2) \times z_j) \text{ at the hidden layer,} \quad (97)$$

Consequently, gain is introduced, and the weight and the bias updates are modified thus, between the hidden layer and output layer, equation (61), (62), (63) and (64) becomes,

$$w_{jk}^1 = w_{jk}^1 + S_{kj} \times a_j^1 \times \text{eta} \times g_k \quad (98)$$

$$w_{jk}^2 = w_{jk}^2 + S_{kj} \times a_j^2 \times \text{eta} \times g_k \quad (99)$$

$$w_{jk}^3 = w_{jk}^3 + S_{kj} \times a_j^3 \times \text{eta} \times g_k \quad (100)$$

the modified updated bias at every control cycle becomes,

$$b_k = b_k + S_{kj} \times \text{eta} \times g_k \quad (101)$$

and equations (72), (73), (74), (75), (76), (77), (78), (79) and (80) become,

$$w_{ij}^{11} = w_{ij}^{11} + S_{ji}^1 \times x(1) \times \eta \times g_k \times g_j \quad (102)$$

$$w_{ij}^{12} = w_{ij}^{12} + S_{ji}^1 \times x(2) \times \eta \times g_k \times g_j \quad (103)$$

$$w_{ij}^{21} = w_{ij}^{21} + S_{ji}^2 \times x(1) \times \eta \times g_k \times g_j \quad (104)$$

$$w_{ij}^{22} = w_{ij}^{22} + S_{ji}^2 \times x(2) \times \eta \times g_k \times g_j \quad (105)$$

$$w_{ij}^{31} = w_{ij}^{31} + S_{ji}^3 \times x(1) \times \eta \times g_k \times g_j \quad (106)$$

$$w_{ij}^{32} = w_{ij}^{32} + S_{ji}^3 \times x(2) \times \eta \times g_k \times g_j \quad (107)$$

and the new bias at every control cycle becomes,

$$b_j^1 = b_j^1 + S_{ji}^1 \times \eta \times g_k \times g_j \quad (108)$$

$$b_j^2 = b_j^2 + S_{ji}^2 \times \eta \times g_k \times g_j \quad (109)$$

$$b_j^3 = b_j^3 + S_{ji}^3 \times \eta \times g_k \times g_j \quad (110)$$

3.3.14 Computing Algorithm

Path planning

Table 8 below shows the waypoint information including the desired surge velocity, x and y-coordinates. The developed guidance system computing algorithm traces a smooth trajectory such that the proposed NN-based autopilot and speed controller can efficiently converge the ship's position to these coordinates, at the desired surge velocity. Figure below shows the desired trajectory generated by the guidance system.

Table 11. Waypoint information

Ship length= 46m			
waypoint	x-positions [m]	y-positions [m]	velocity [m/s]
0	0	0	0
1	1800	1800	6
2	3600	1800	6
3	5400	0	6
4	3600	-1800	6
5	900	-1800	6
6	0	0	0

Figure 12 shows the smooth trajectory generated by the guidance system. This confirms that the guidance system algorithm suffices for the proposed control system.

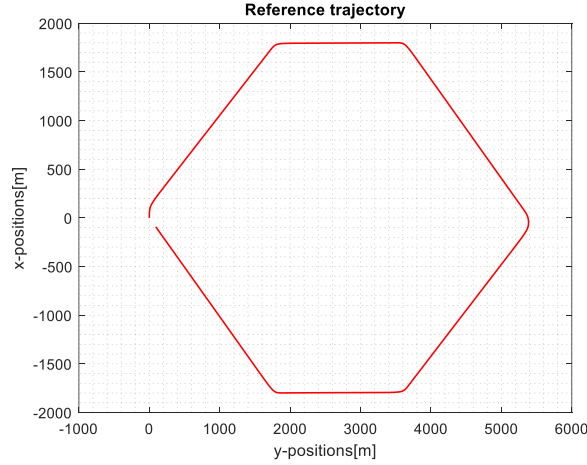


Figure 36. Predefined trajectory generated by the guidance system.

To effectively represent and simulate the unberthing, trajectory tracking and berthing missions of a surface vessel together, a set of 17 waypoints are carefully selected. Waypoints number (1) to (4) represent the unberthing mission, waypoint numbers (4) to (14) represents a round-trip trajectory tracking mission and waypoint numbers (14) to (17) represents the berthing mission.

The x and y-coordinates in the earth-fixed frame are: [(0,0), (50,700), (400, 1400), (400, 2800), (500, 3200), (1000, 3600), (1500, 3200), (1600, 2800), (1600, 0), (1600, -2800), (1500,-3200), (1000, -3600), (500, -3200), (400,-2800), (400, -1400), (0,-700), (0,0)]. The surge velocities for each way point in m/s are: [0,4,4,4,4,4,4,4,4,4,4,4,4,3,2,1]. Simulation of the guidance system yields the reference trajectory as shown in Figure 13 with outputs: $(U_d(t), y_d(t), x_d(t), \psi_d(t))$. These outputs are the setpoint for NN-controller cascade.

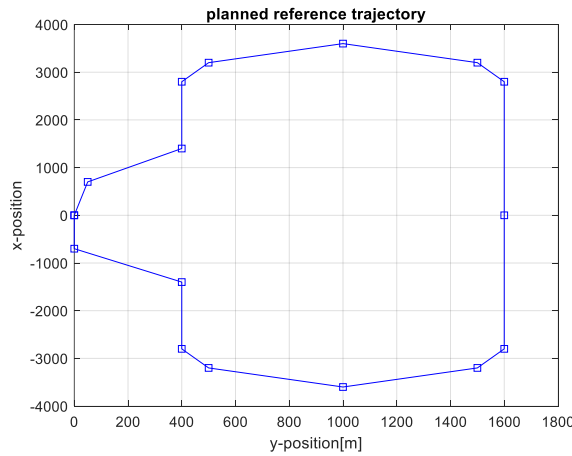


Figure 37. Planned reference trajectory

The Neural Network-based trajectory control of the Shioji Maru

This research proposes the approach developed by Nguyen (2007), where predetermined waypoints and speed (reference trajectory) are used as a setpoint to control the ship's heading and speed. The guidance system generates the reference trajectory, the ANN-based autopilot makes

the course control decisions and guides the ship rudder for course changing through the reference trajectory. The NN-based speed controller gives the speed control information to the controllable pitch propeller of the vessel to maintain the set speed over the reference trajectory. The position controller is used to minimize cross track error during trajectory tracking. The algorithm is developed such that it suffices for vessels with single thruster (Bow thruster). As suggested by Lekkas (2014), the position error for the vessel position (x, y) in vessel-fixed frame is given by,

$$\begin{bmatrix} x_e \\ y_e \end{bmatrix} = R^T(psi) \begin{bmatrix} x - x_t \\ y - y_t \end{bmatrix} = \begin{bmatrix} \cos(psi) & -\sin(psi) \\ \sin(psi) & \cos(psi) \end{bmatrix} \times (psi) \begin{bmatrix} x - x_t \\ y - y_t \end{bmatrix} \quad (111)$$

Where R^T is a transformation matrix and x_e and y_e are the earth-fixed position errors of the ship during trajectory track. Combining the position errors in both x and y earth-fixed reference frame gives the cross-track error as,

$$e_c = (x - x_t) \cos(psi) + (y - y_t) \sin(psi). \quad (112)$$

To minimize the position error, e_c and the derivative $\dot{e}_c$ is selected as an input to the NN-based position controller. The computing algorithm is illustrated in Figure 14 and the controller parameters are shown on Table 9. The simulation studies have been conducted using the mathematical model of Shioji Maru and with control programs developed in the MATLAB and LabVIEW software environment.

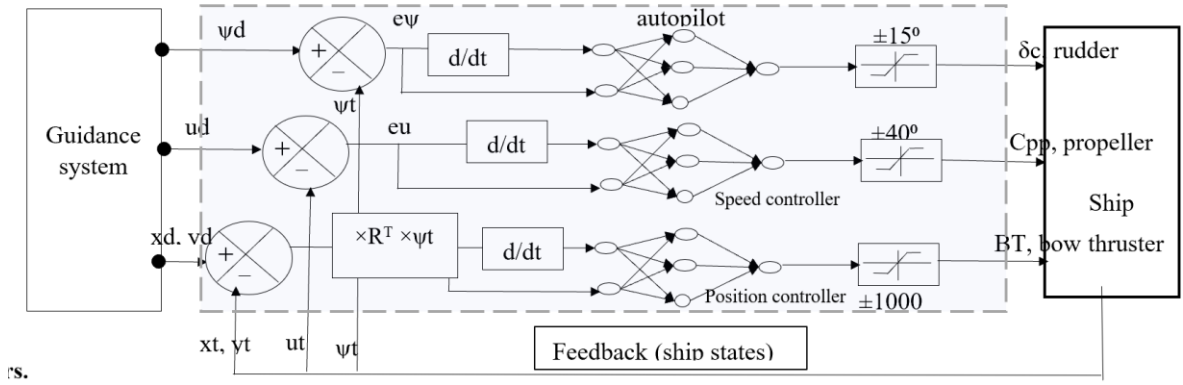


Figure 38. Illustration of the computing algorithm

Table 12 Controller parameters

PID [Kp, Ki, Kd]			ANN-based controllers			
autopilot	Speed controller	Position controller		autopilot	Speed controller	Position controller
[93 2 0.09]	[100 0.01 0.15]	[12 0.001 0.15]	Learning rate	0.001 (0.01 if $e\psi > 10^\circ$)	0.00001	0.8
			Number of layers/neurons	3/3	3/3	3/3

3.3.15 low speed maneuvering (berthing and unberthing mission)

Berthing and unberthing mission require that ships maneuver at very low speeds. At such low speeds, the propeller rpm (speed) is very low that the effective water flow over the rudder is greatly reduced and consequently, the rudder's ability to steer the ship is minimized.

Simulation studies will be conducted to investigate how the NN-based autopilot and speed controllers perform for an automatic berthing and unberthing scenario at very slow ship speed.

3.3.16 investigation of the ship's turning circle at very low speed

To evaluate the suitability of the low-speed mathematical model of Shioji Maru for slow speed berthing and unberthing missions, simulations are conducted for several scenarios.

In figure 15A, the simulation results show that at 20° commanded rudder angle (without the thrusters), the ship maneuvers slowly and the turning circle is very large. In 15B, the rudder angle is turned to -20° and the simulation results show the ship making a turning circle in the opposite direction. In 15C and 15D, the thrusters are added to aid the maneuvering and the turning circles become relatively smaller than that in 15A and 15B.

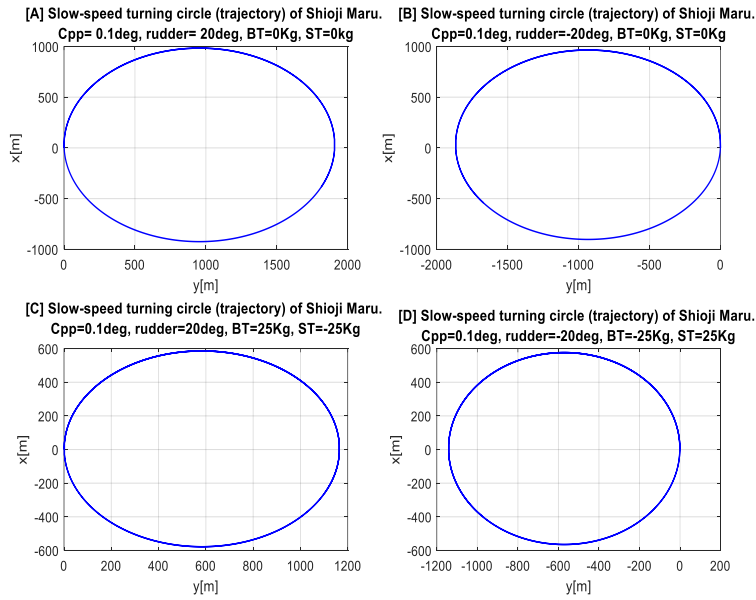


Figure 39. Turning Circles for slow speed model of Shioji Maru.

The reference trajectory has been defined to reflect the berthing and unberthing trajectories with the surge velocity set at 1m/s.

3.3.17 modifications to the design algorithm for berthing and unberthing scenario.

In the design of the guidance system for slow speed maneuvering (berthing or unberthing), the circle of acceptance, R_o for turning at every waypoint has been selected as,

$$R_o = L_{pp}$$

Considering the very low speed maneuvering, the smaller circle of acceptance has been selected due to the slow transition at the waypoints.

4.0 RESULTS AND SIMULATION

4.1 COMPARISON STUDY

The autopilot controller is tested without a speed controller at a constant maximum cpp of 15° and a desired heading angle of 30° . While the PID-based autopilot attained the heading setpoint in 140s, the NN-based autopilot shows a faster response and reached the setpoint in 100s. The plot of the ship heading for the two controllers are shown in Figure 16. NN-based autopilot shows desirable characteristics over the PID-based autopilot in terms of the rise and settling time. This is an important characteristic particularly suited for trajectory tracking where a quick response is desired to minimize the tracking error. The results are compared as shown on Table 10.

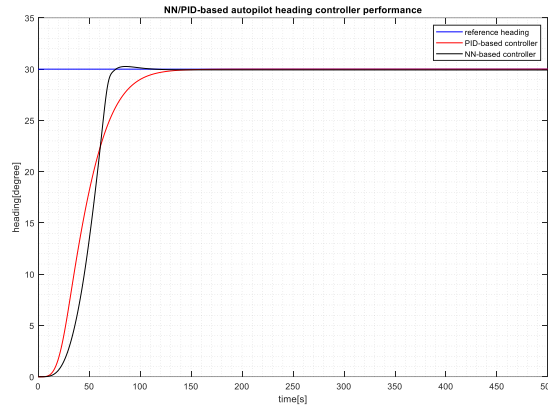


Figure 40. Plot of heading angle by NN-based and PID-based autopilot (performance evaluation)

PID controller gain, $K_s = [0.99 \ 0 \ 22.9]$.

NN parameters, weights and bias,

weights at the first layer, $w_1 = [-30.5 \ -5; 55 \ 30; 25 \ 85]$

weights at the 2nd layer, $w_2 = [0.39 \ 0.066 \ 0.48]$.

Bias at first layer, $b = [-0.54 \ 0.9 \ 0.38]$.

Bias at the output layer, $b_4 = 0.000001$.

Learning rate, $\eta = 0.000048$.

Table 13. Comparing the NN-based autopilot against the PID-based autopilot.

	PID-based autopilot	NN-based autopilot
Rise time [s]	135	80
Settling time [s]	135	105
Over shoot	0%	0.67%

Peak value [m/s]	30°	30.2°
Steady state error [m/s]	±0.000001	±0.000001
Closed loop stability	stable	stable

Evaluation of the speed controllers at a constant heading shows that both the NN-based controller and PID-based controller had similar performance. The simulation results of the speed controllers are shown on Figure 17.

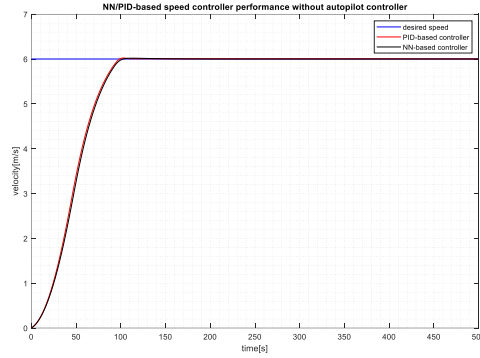


Figure 41 Plot of surge velocity control by NN-based and PID-based speed controller (performance evaluation)

PID controller gain, $K_{shp} = [20 \ 0.044 \ 220]$.

NN parameters, weights, and bias.

weights at the first layer, $w_{u1} = [1.02 \ 1.02; 1.04 \ -1.02; 1.02 \ 1.01]$

weights at the 2nd layer, $w_{u2} = [0.05 \ 0.05 \ 0.04]$.

Bias at hidden layer, $b_u = [0.202 \ 0.202 \ 0.205]$.

Bias at the output layer, $b_{u4} = 3.3$.

Learning rate, $\eta = 0.000698$.

Table 14. Comparing the NN-based speed controller against the PID-based speed controller.

	PID-based speed controller	NN-based speed controller
Rise time [s]	100	100
Settling time [s]	100	100
Over shoot	0%	0%
Peak value [m/s]	6	6
Steady state error [m/s]	±0.000001	±0.000001
Closed loop stability	stable	stable

The combination of the autopilot and speed controller is evaluated for both the NN-based controller design and the PID-based controller design as shown in Figure 18. The respective autopilots are set at 30° desired heading and the speed controllers set at 6m/s surge velocity. The simulation results show that the NN-based autopilot has a faster response relative to the PID-based autopilot and in speed control, both controllers performed similarly.

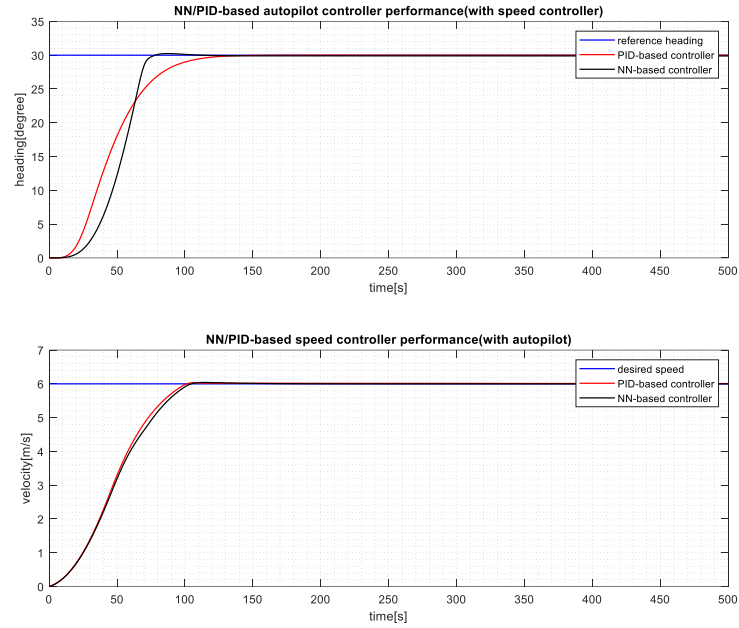


Figure 42. Combined NN-based autopilot and speed controller Vs combined PID-based autopilot and speed controller.

4.2 COMPARISON STUDY AT LOW SPEED MANEUVERING

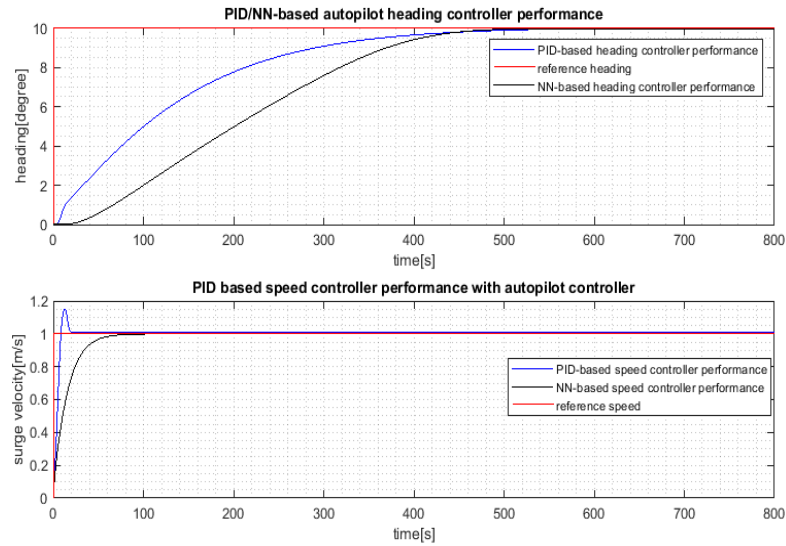


Figure 19. Performance of the controllers at slow speed maneuvering.

Figure 19 shows the simulation results which aims to compare the performances of the NN-based controllers and the PID-based controllers at very low-speed berthing scenario. The NN-based autopilot shows a slightly faster response at 1m/s surge velocity.

In speed control, The PID-based controller exhibited an uncontrollable overshoot whereas the NN-based speed controller chased the setpoint steadily without any overshoot. Unlike the PID-based speed controller which has a steady state error, the NN-based controller exhibited no steady state error.

4.3 POSITION ERRORS AT DIFFERENT GAIN VALUES.

From Figure 20, it is observed that at a lower gain value, there was a reduced oscillation and a lower position error

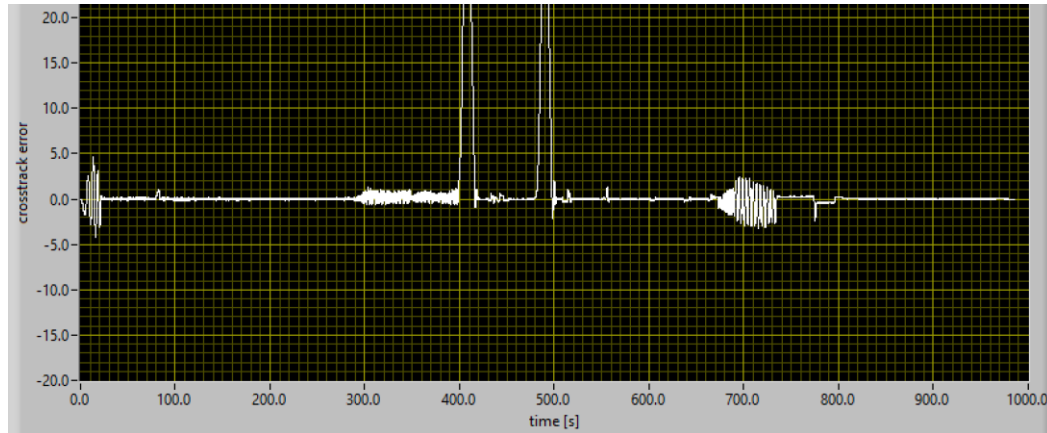


Figure 20a. Position error at gain =0.5

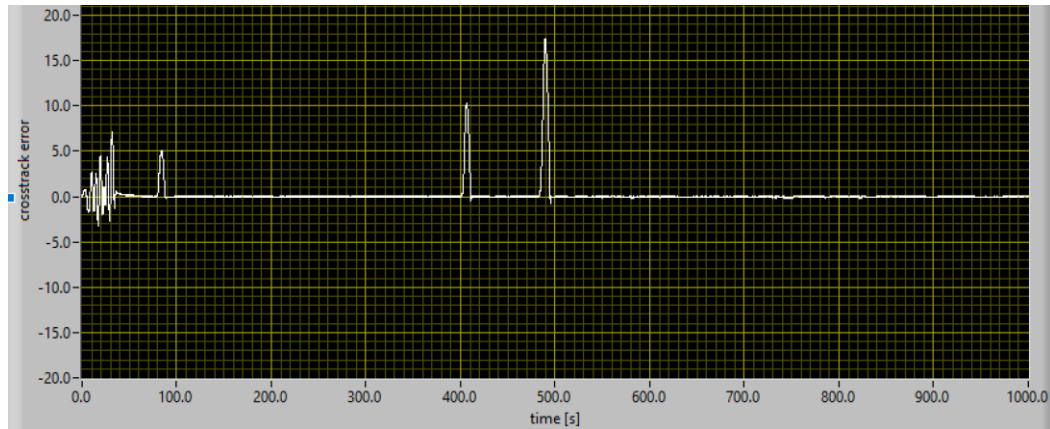


Figure 43b. Position error at gain=0.1

4.4 SPEED CONTROL TEST

To evaluate how the speed controllers will perform in a berthing scenario, a speed control simulation test is carried out. The set speed of the ship is increased to 4m/s then gradually reduced (as in a berthing scenario) from 4m/s through to 0m/s. (from 4m/s to 1m/s to 0m/s). The response of the controllers to the speed changes are given in Figure 21. Both the PID-based and NN-based

controllers showed almost the same rise time, but the PID-based controller (in blue colour) has a slow response and a continuous steady state error during the speed reduction stage. This suggests the susceptibility of the PID-based speed controller to possible along-track errors during berthing control. From the results, it can be deduced that the NN-based speed controller has a better performance than the PID-based speed controller for automatic ship berthing.

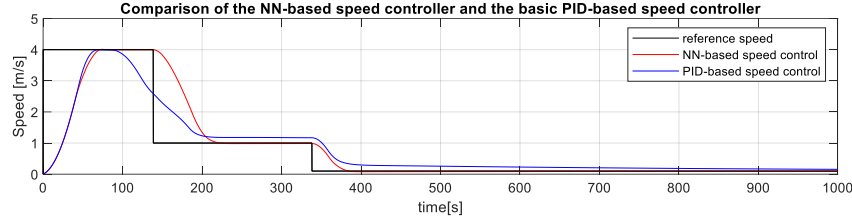


Figure 44. simulation studies on speed control for NN-based and PID-based controllers.

4.5 COURSE KEEPING AND COURSE CHANGING CONTROL

Similar comparison studies on the PID-based autopilot and the NN-based autopilot is shown in Figure 22. The set heading of the ship is changed from 0° to 20° , to 30° and to 45° over the entire simulation time. This study is aimed at finding out how fast the controllers will respond to course changes during trajectory control. The result reveals that the NN-based autopilot has a faster rise and settling time compared to the PID-based autopilot. These suggests that the PID-based autopilot is likely to cause more cross track errors compared to the NN-based controllers. Hence, the NN-based autopilot can be concluded to have a better performance in a course changing and course keeping

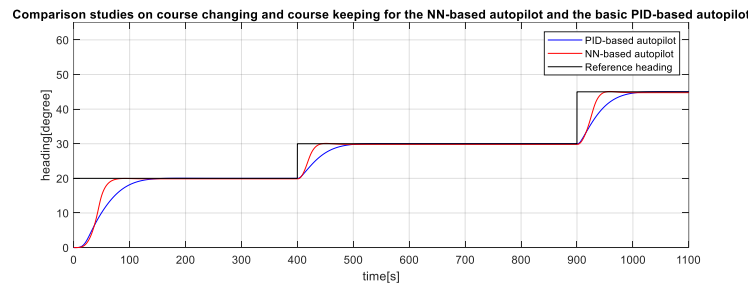


Figure 45 simulation studies on the performance the NN and PID-based controllers during course change

4.6 SIMULATION STUDIES ON BERTHING CONTROL.

Simulation studies is carried out to ascertain how the controllers will perform in a berthing mission of a ship, from the Tamar river to the port of Bell Bay in Northern Tasmania. The berthing trajectory is planned as shown in Figure 23. Simulation results as shown in Figure 23 reveals that the cross-track error of the ship when the PID-based controllers are in use and the cumulative effect of the PID speed control errors led to an along-track error at the final berthing point. Figure 23 reveals the near-perfect berthing trajectory of the ship (in red) when using NN-based controllers. The slight oscillations at the start of the berthing can be attributed to the initial rise time of the speed controller and the fast response of the position controller trying to maintain the set course.

The result shows that the NN-based controllers outperformed the PID-based controllers in a berthing scenario.



Figure 46. Aerial image of the planned berthing trajectory. (Google 2020)

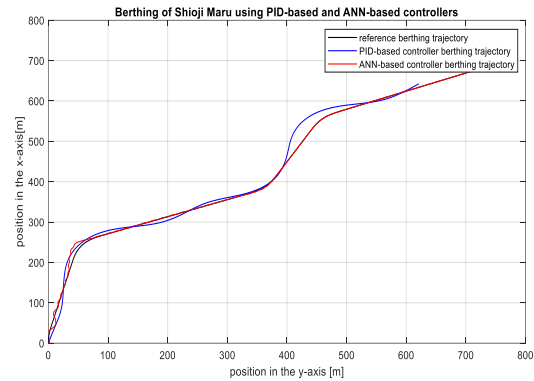


Figure 47. Berthing trajectory of Shioji Maru using PID-based and NN-based controllers.

4.7 SIMULATION STUDIES ON FULL MISSION TRAJECTORY TRACKING CONTROL

Figure 25a shows the plot of trajectory tracking control of the Shioji Maru using different sets of controllers. The trajectory tracking control plot using PID-based controllers shows multiple tracking errors at the unberthing stage, turning points and at the berthing stage. This highlights the slow response of the PID-based controllers when the setpoints are changed. At the unberthing stage, the slow response of the PID-based speed controller caused a large tracking error and the PID-based autopilot also showed similar slow responses at the turning points. Similarly, a slight along-track error is observed at the berthing stage which prevented the ship from reaching the last waypoint. Unlike the PID-based controllers, the ANN-based controllers exhibited a near-perfect tracking response regardless of changes in waypoints (turning points).

Figure 25b shows the plot of the position error of the respective set of controllers and this highlights the poor response of the PID-based controllers even further. Through the trajectory tracking, the maximum position error shown by the ANN-based controllers is +22m over maximum interval of 10 seconds before attaining a steady state with zero error. The PID-based controllers showed multiple tracking errors with maximum position error of +50m in an interval of 100seconds. Thus, it can be concluded that the ANN-based controllers have a better performance than the basic PID-based controllers in a trajectory tracking control for a surface vessel. Similar results can also be observed on a different trajectory as shown in Figure 26.

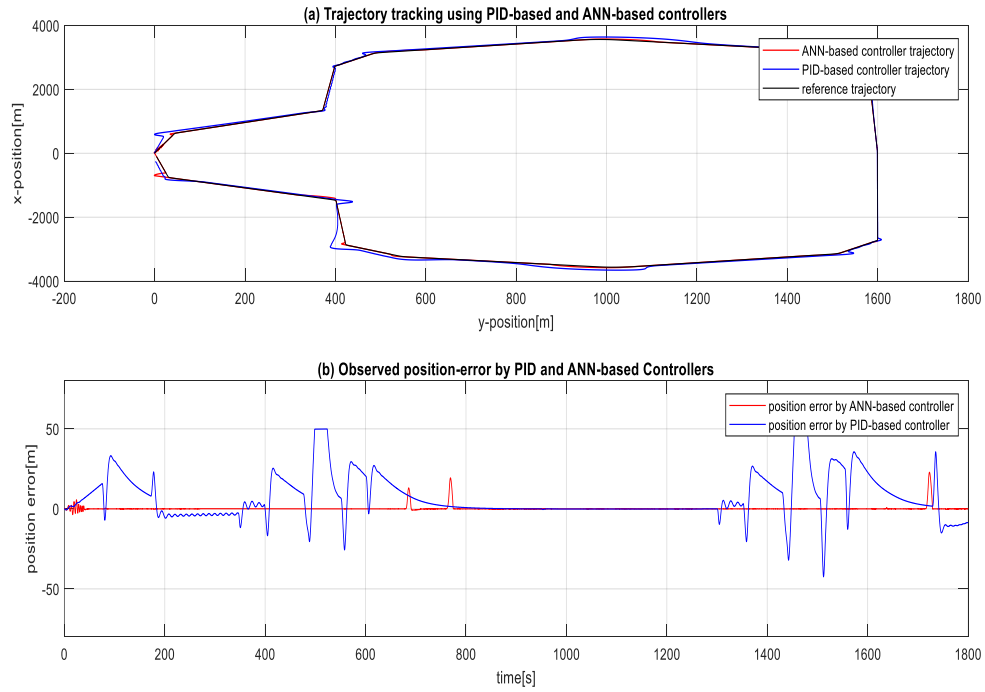


Figure 48. Plot of full mission trajectory tracking of Shioji Maru vessel. (b) Plot of the ship's position-error over time.

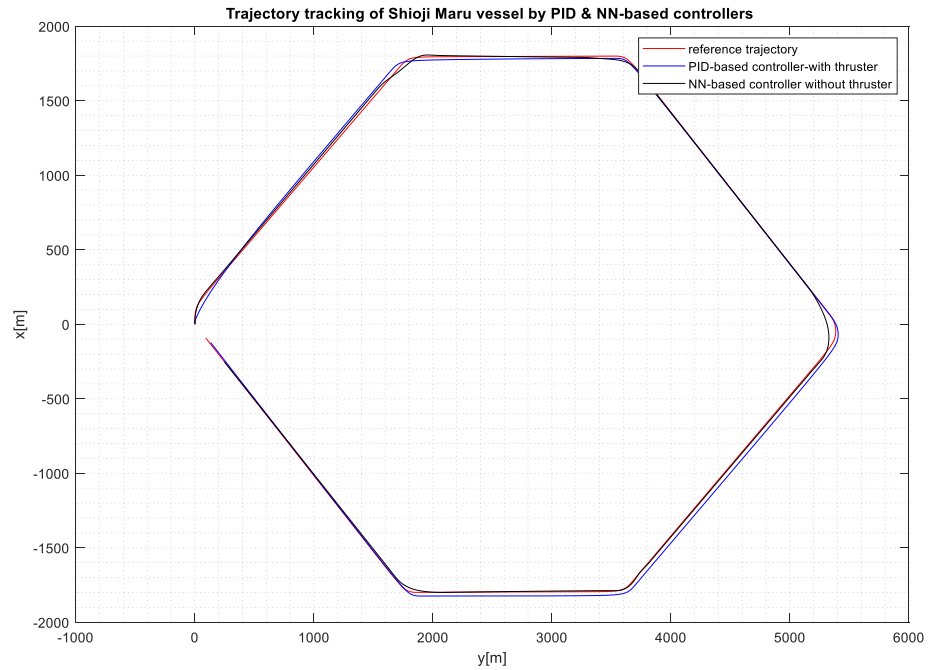


Figure 49. Trajectory tracking using NN-based controllers with the PID-based controllers used as a benchmark and for comparison.

5.0 EXPERIMENTAL PLAN

The aim

The project experiment will seek to validate the designed control algorithm for trajectory tracking, berthing and unberthing missions for Bluefin. The experimental results and performance of the control algorithm will be compared against the results obtained from the numerical simulation studies.

a) The scope of experiments

The scope of the experiment will entail a confirmation that the performance level of the NN-based control algorithm suffices in making control decisions required for trajectory tracking, berthing and unberthing missions.

b) The objectives of experiments

The experiments will be conducted to validate the results of the numerical simulations for the development of a neural network-based control system for trajectory tracking, berthing and unberthing missions.

c) Expected Outcomes:

Results of the experiments should validate the results of the numerical simulations

To run the neural network-based algorithm successfully

To successfully collect good data for documentation and reporting

d) Choice of factors, levels, and ranges

The dynamic system (Bluefin) should provide the following data:

Propeller RPM (500 RPM to 900 RPM)

Rudder angle (deg) (-40 to +40 deg)

Digital compass: heading/yaw angle 0 to 360 degrees.

Velocities: u and v 0 to 0.8m/s

Position in x-axis and y-axis

e) Choice of design

For verification of performance of designed control algorithm; the control algorithm will be tested under the following cases,

Trajectory tracking by Neural network controller

Speed control by neural network controller

f) Location of experiment

Experiments will be conducted in Trevallyn Lake on two sunny days.

g) Experimental activities

Investigation of the turning circle maneuvering capability of the ship model

Investigating the autopilot performance

Investigating the speed controller performance

High speed trajectory tracking

Slow speed berthing scenario

h) Post experimental data processing, analysis and conclusion

To process the raw data from the low-cost sensors, the following algorithm are needed,
signal conditioning

filters (low pass and Kalman filters)

Drawing conclusions, and making recommendations

i) Facilities, tools and support needed

Any available technician, vehicle to carry the model scaled boat and all experiment hardware
Tools.

A colleague for assistance

A laptop

Technical Support/Assistance from Lab technician.

5.1 LOCAL COORDINATES CALCULATION REQUIRED WHEN USING THE GPS HARDWARE.

Experiments to validate the numerical simulations are carried out in Tasmania, Australia where the local coordinates are 42.029423°S and 146.5904°E.

To use the GPS hardware which takes its origin from the prime meridian, the readings from the GPS are offset to reflect the location where the experiment is conducted.

Distance between Latitude of Tasmania and the prime meridian.

Considering that Tasmania is located at latitude 42.029423°S the distance between the prime meridian with latitude 0° and Tasmania (42.029423°S) is determined thus:

Note: Distances to the North are considered positive, distances to the south are considered negative.

Distances eastwards are considered positive, westwards are negative.

1°= 60 minutes

Total minutes between prime meridian (0°) and latitude of Tasmania (42.029423°S) =

$$42.029423 \times 60 = 2,521.76538 \text{ minutes}$$

Distance between 1/60° or 1 minute of latitude along any line of longitude is defined as.

1 minute=1852 metres,

therefore.

$$\begin{aligned} 2,521.76538 \text{ minutes} &= (2521.76538 \times 1852) \text{ metres} \\ &= -4,670,309.48376 \text{ metres.} \end{aligned}$$

Distance between Longitude of Tasmania and the prime meridian.

Distance between 1° of longitude at any given latitude is expressed as;
 $\cos(\text{Latitude}) \times \text{length between } 1^\circ \text{ of longitude at the equator.}$

$$\text{length between } 1^\circ \text{ of longitude at the equator} = 11321 \text{ metres}$$

$$\begin{aligned} \text{Distance between the prime meridian and the longitude of Tasmania (146.5904}^\circ) \\ = 146.5904 \times \cos(42.029423) \times 11321 \text{ metres} \end{aligned}$$

$$= 12121448.5913 \text{ metres.}$$

Hence, distance readings numerically obtained from the experimental GPS coordinates will be offset by the distances between the prime meridian and Tasmania.

6.0 CONCLUSIONS

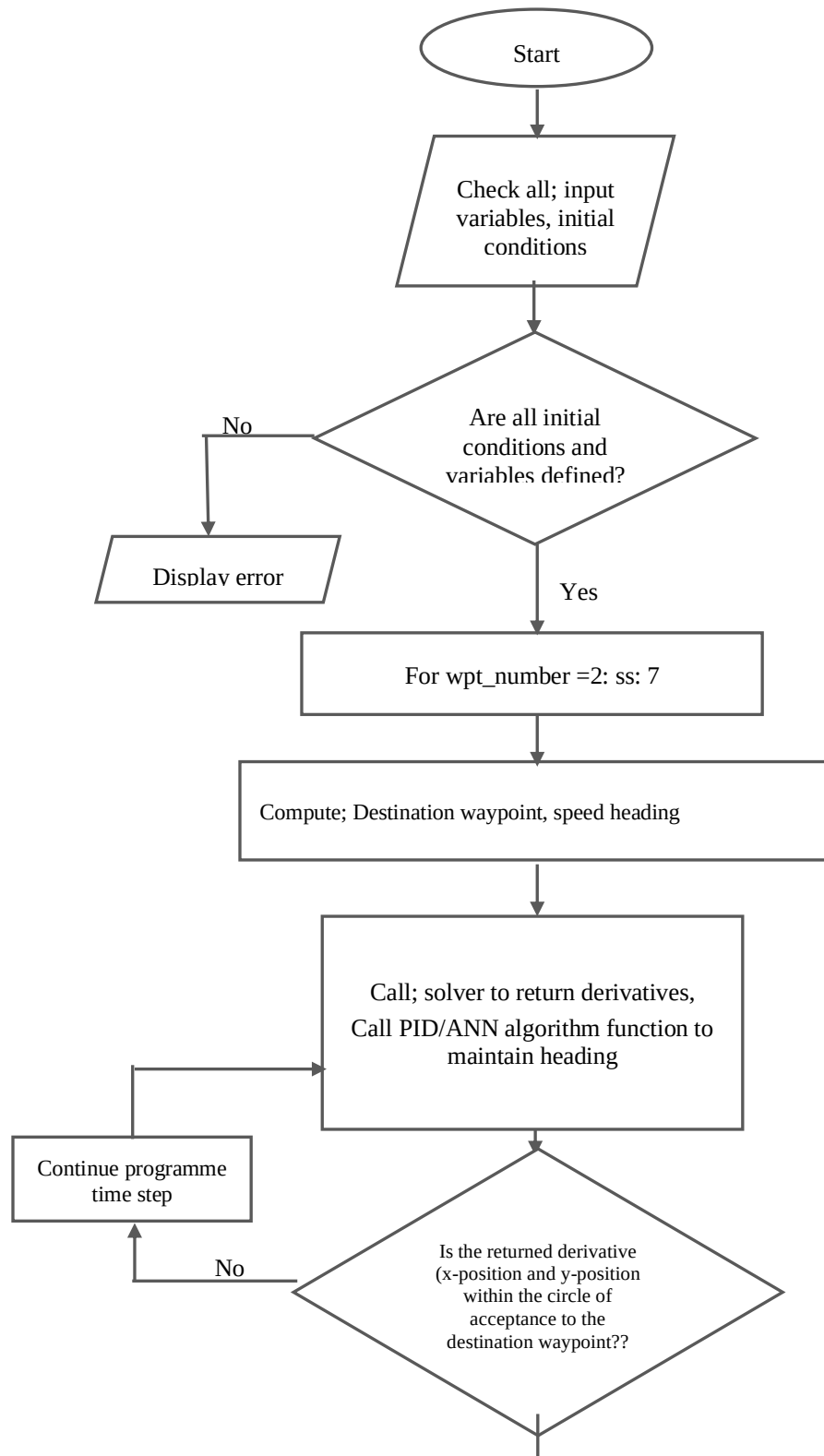
This research work covered the development of a neural Network-based control system for trajectory tracking, berthing and unberthing mission of a ship. The developed marine control system consists of the guidance system, the autopilot, the speed controller, and the position controller. The guidance system used a predefined waypoint information to generate the reference trajectory information for the controllers. In the guidance system algorithm, the line of sight algorithm was used to compute the desired heading, the PID control law was used for course keeping and the linear ship model (T-K model) was used to generate the reference trajectory consisting of the desired ship states. The NN-based controllers for control decision making were developed using the multilayer perceptron with the backpropagation learning technique. Comparison studies were carried out between the NN-based controllers and the PID-based controllers. The results show that the NN-based controllers have better performance than the basic PID-based controllers. Unlike the PID-based controllers, the NN-based controllers have faster response and an ability to adapt to changes in course dynamics. Positive results from research studies suggest that the neural network-based controllers could be the future of advanced marine control systems and this could become the solution to a more efficient ship trajectory tracking control, automatic berthing and unberthing in shipping ports all over the world. The outbreak of COVID-19 prevented the author from validating the results by experiments and hence, future researchers will see to improve the learning ability of the NN-based control algorithm and to validate the simulation results experimentally.

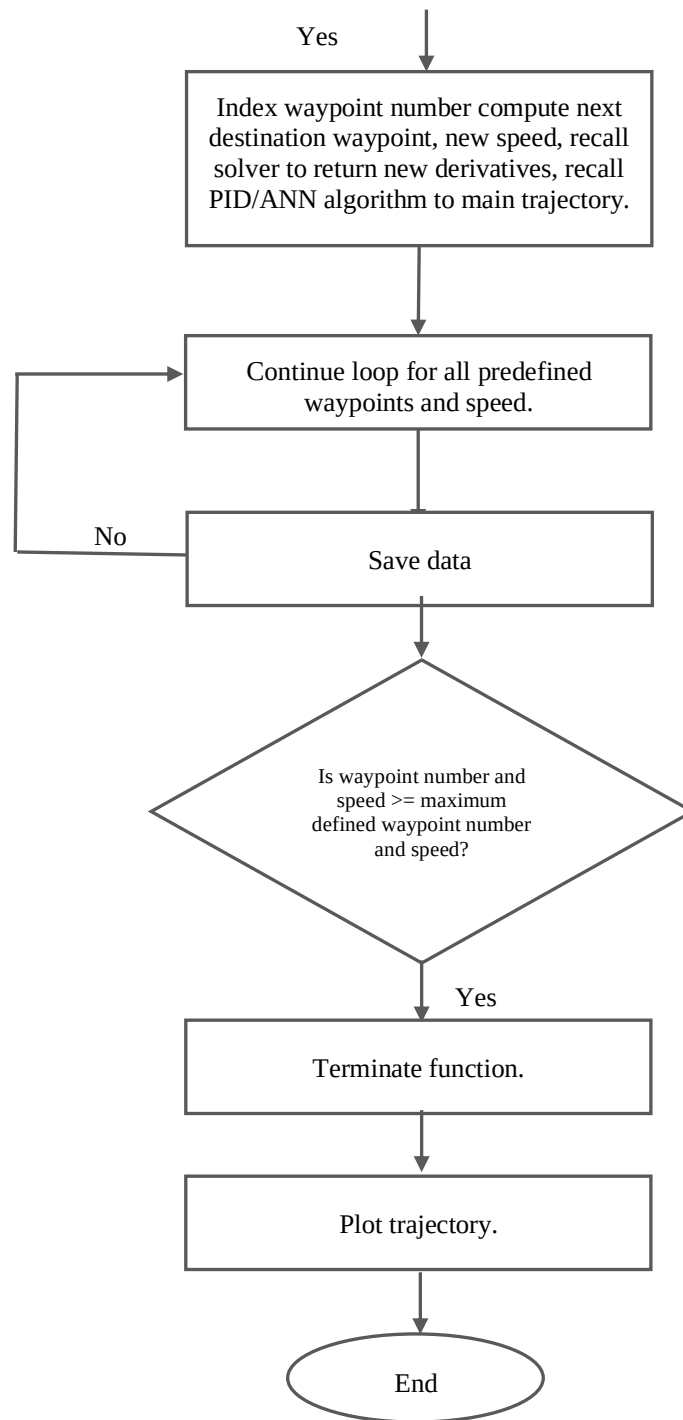
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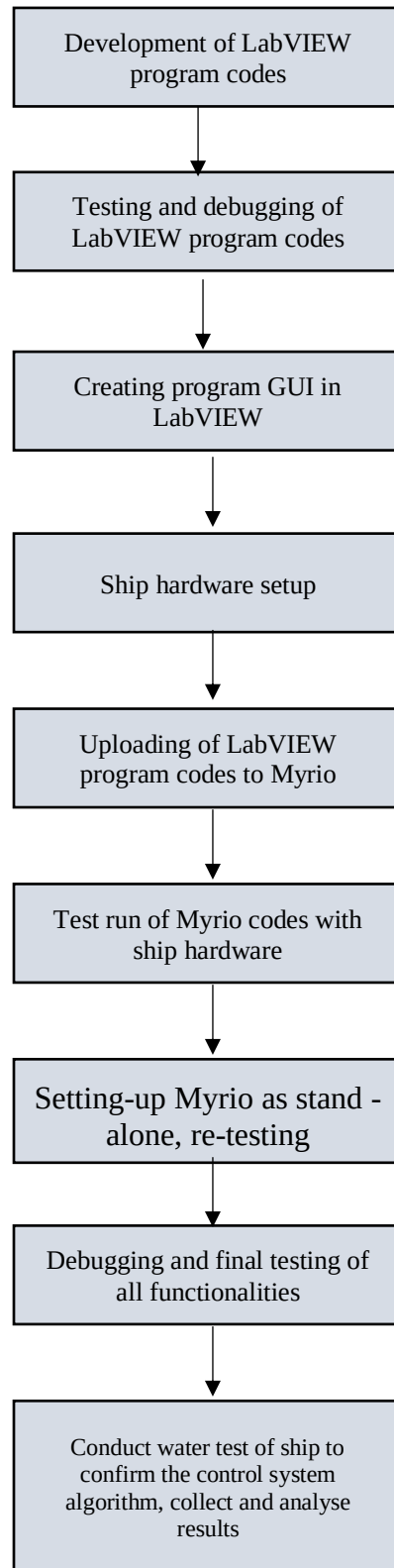
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APPENDIX A: PROGRAM FLOW CHART





APPENDIX B: EXPERIMENTAL DESIGN FLOW CHART



APPENDIX C:

HYDRODYNAMIC AND DERIVATIVE COEFFICIENTS OF SHIOJI MARU

Items	Symbols	Values	Units
Mass of ship	m	6.7418×10^4	$[kgs^2/m]$
Added mass of ship in x-direction	m_x	2.6710×10^5	$[kgs^2/m]$
Added mass of ship in y-direction	m_y	5.4032×10^4	$[kgs^2/m]$
Z-axial inertial moment & added one	$I_{ZZ} + J_{ZZ}$	1.5184×10^7	$[kgs^2m]$
Length between perpendiculars	L_{PP}	46.0	$[m]$
Mean draft	d	2.85	$[m]$
Immerged area of ship hull	S_W	480.0	$[m^2]$
Ship resistant coefficient	C_t	4.32750×10^{-3}	$[-]$
Resistance deduction coefficient	t_R	0.215	$[-]$
Effective wake factor at the rudder position	W_R	0.22	$[-]$
Projected area of the rudder	A_R	4.25	$[m^2]$
X-coordinate of the rudder sharp	x_R	-22.649	$[m]$
Point at which the intervening force of rudder is acting on the ship's hull	x_H	-14.4	$[m]$
Coefficient due to mutually intervening between the ship's hull and rudder	a_H	0.219	$[-]$
Slope due to inclined angle of single rudder normal pressure	f_a	1.85	$[-]$
Adjusting factor 1	γ_R	0.4998	$[-]$
Adjusting factor 2	l_R	-0.77735	$[-]$
Velocity deduction factor at rudder position	k_x	0.6177	$[-]$
Propeller diameter	D_P	2.2	$[m]$
Propeller revolution per second	n	5.0	$[1/s]$
Thruster deduction coefficient	t_P	0.193	$[-]$
Effective wake coefficient at propeller position	W_P	0.28989	$[-]$
Propeller pitch corrected constant	C_{T0}	20.86	$[deg]$
Experimental propeller thruster coeff. 1	C_0	3.509177×10^{-1}	$[-]$
Experimental propeller thruster coeff. 2	C_1	2.291329×10^{-2}	$[-]$
Experimental propeller thruster coeff. 3	C_2	-3.064803×10^{-1}	$[-]$
Experimental propeller thruster coeff. 4	C_3	-5.197373×10^{-3}	$[-]$
Experimental propeller thruster coeff. 5	C_4	3.784367×10^{-4}	$[-]$
Experimental propeller thruster coeff. 6	C_5	1.681795×10^{-2}	$[-]$
Experimental propeller thruster coeff. 7	C_6	-3.186904×10^{-6}	$[-]$
Experimental propeller thruster coeff. 8	C_7	2.015531×10^{-3}	$[-]$
Experimental propeller thruster coeff. 9	C_8	2.272834×10^{-6}	$[-]$
Experimental propeller thruster coeff. 10	C_9	-1.803216×10^{-1}	$[-]$
Derivative of hydrodynamic force to surge	X'_v	-0.111418×10^{-1}	$[-]$
Derivative of hydrodynamic force to surge	X'_{vr}	0.176915	$[-]$
Derivative of hydrodynamic force to surge	X'_r	0.137031×10^{-2}	$[-]$
Derivative of hydrodynamic force to surge	X'_{vv}	0.955500×10^{-2}	$[-]$
Derivative of hydrodynamic force to surge	X'_{rr}	0.181790×10^{-3}	$[-]$
Derivative of hydrodynamic force to sway	Y'_v	-0.28689	$[-]$
Derivative of hydrodynamic force to sway	Y'_{vr}	-0.136844	$[-]$
Derivative of hydrodynamic force to sway	Y'_r	-0.355849×10^{-2}	$[-]$
Derivative of hydrodynamic force to sway	Y'_{vv}	-0.554027	$[-]$
Derivative of hydrodynamic force to sway	Y'_{rr}	0.128952	$[-]$

Derivative of hydrodynamic force to yaw	N'_v	-0.140762	[-]
Derivative of hydrodynamic force to yaw	N'_{vr}	-0.169032	[-]
Derivative of hydrodynamic force to yaw	N'_r	-0.612311×10^{-1}	[-]
Derivative of hydrodynamic force to yaw	N'_{vv}	-0.974819×10^{-1}	[-]
Derivative of hydrodynamic force to yaw	N'_{rr}	0.117956×10^{-1}	[-]
Time constant of rudder	T_{RUD}	11.9	[s]
Time constant of propeller pitch	T_{CCP}	2.8571	[s]
Lateral projected area	A_{of}	58.10	[m ²]
Transverse projected area	A_{os}	275.0	[m ²]
Approximation of wind pressure coeff. C_X	C'_X	-0.322456	[-]
Approximation of wind pressure coeff. C_Y	C'_Y	-0.951717	[-]
Approximation of wind pressure coeff. C_N	C'_N	-0.108088	[-]
Acceleration of gravity	g	9.81	[m/s ²]
Salty Water Density (at 20°C)	ρ_w	1024.851	[kg/m ³]
Air Density (at 20°C)	ρ_a	1.2263	[kg/m ³]

Table A2.2 Derivative coefficients of *Shioji Maru* (high speed)

Speed [knot]	u [m/s]	Y_v	Y_r	N_v	N_r
-1.4	-0.7202222	-0.035567	-0.179128	0.013795	-0.280939
0.0	0.0	0.0	0.179128	0.0	-0.280939
1.4	0.7202222	-0.035567	0.179128	-0.013795	-0.280939
3.0	1.5433333	-0.334914	0.036669	-0.129903	-0.057510

Table A2.3 First order dimensional derivatives of *Shioji Maru* (low speed)

APPENDIX D: REVISED/UPDATED MILESTONE

Task Name	Duration	Start	Finish
PROJECT INITIATION	77 days	25/02/19	11/06/19
Project handover from Project coordinator and supervisor	2 days	25/02/19	26/02/19
Theoretical studies on research background	95 days	27/02/19	9/07/19
Developing a Project Plan	18 days	1/03/19	26/03/19
Submitting Version 1 of project plan to Project supervisor for review and comments	3 days	27/03/19	29/03/19
Revision and submission of final version of project plan	4 days	27/03/19	01/04/19
Further Literature review/background theoretical studies.	51 days	2/04/19	11/06/19
NUMERICAL SIMULATIONS	82 days	2/04/19	31/11/19
Improving programming skills	71 days	2/04/19	09/07/19
Numerical simulations for verification of the open loop characteristics of ship model	6 days	10/04/19	17/04/19
Formulation of algorithm 1: waypoint generator.	11 days	15/04/19	29/04/19
Formulation of algorithm 2. autopilot with PID controller.	28 days	29/04/19	05/06/19
i) Finalising interim report, ii) hand over of interim report to supervisor for review.	4 days	5/06/19	10/06/19
Submission of interim report.	2 days	10/06/19	11/06/19
iterations and refinement of formulation algorithm 1 and 2	17 days	12/06/19	4/07/19
Mathematical modelling of the PID-based autopilot system	2 days	5/07/19	8/07/19
developing MATLAB codes for the waypoint generator, PID-based autopilot system and PID-based speed controller.	2 days	8/07/19	9/07/19
Numerical Simulations to verify performance and robustness of the PID-based autopilot system.	2 days	8/07/19	9/07/19
Combining the waypoint generator, autopilot, ship controller and ship model	2 days	8/07/19	9/07/19
i) Numerical Simulations to verify performance of the PID-based control system for trajectory tracking.	108 days	8/07/19	4/12/19
i) Collection of results, ii) documentation (PID-based autopilot and controllers).	2 days	8/07/19	9/07/19
i) Creating an experimental plan, ii) risk assessment for experiments.	2 days	8/07/19	9/07/19
Background theoretical studies on ANN methodology	15 days	26/06/19	11/07/19
formulating an algorithm to develop an ANN-based controller	45 days	26/06/19	10/08/19
Writing MATLAB codes for the ANN-based controller	30 days	11/08/19	10/09/19
Numerical Simulations to verify performance of the ANN-based control system.	20 days	10/09/19	01/10/19
Modification of ANN-based algorithm for performance optimization.	9 days	1/10/19	10/10/19
Final numerical Simulations to verify performance of the ANN-based control system.	2 days	11/10/19	12/10/19
Documenting technical details of the ANN controllers	10 days	12/10/19	22/10/19
Submission of interim report 2 to supervisor for review	2 days	22/10/19	23/10/19
Submission of interim report 2.	1 day	29/10/19	29/10/19

Development of a neural network-based control system for trajectory tracking, berthing and unberthing missions.

Completion of numerical simulations for slow speed berthing and unberthing.	31 days	30/10/19	30/11/19
i)Improving software skills for LabVIEW, ii) understanding the hardware required for the project.	31 days	30/10/19	30/11/19
PREPARATION FOR EXPERIMENTS AND EXPERIMENTS.			
Testing bluefin open-loop system, developing LabVIEW program codes for guidance system.	9 days	24/02/20	03/03/20
developing LabVIEW program codes for PID-based autopilot and speed controller	3 days	03/03/20	5/03/20
Developing and replacing PID with ANN in the autopilot and speed controller.	16 days	5/03/20	17/03/20
Adding program codes for GPS, Compass, GPS-RTK.	8 days	17/03/20	24/03/20
Finalising date for experiments.	1 day	24/03/20	24/03/20
Uploading programs to Myrio and testing program codes for; Propeller, rudder, GPS, Compass and GPS-RTK on the Bluefin hardware	8 days	24/03/20	31/03/20
Debugging and improvements	8 days	31/03/20	7/04/20
Further program tests and verification of all functionalities.	2 days	7/04/20	8/04/20
Documenting details of the hardware and software setup for experiments	31 days	24/02/20	28/04/20
Final preparations and Pre-experimental checks. i) Hardware checks ii) Software and program code checks	3 days	8/04/20	10/04/20
DAY 1: i) Experiments at the Lake. ii) collection of results	1 day	12/04/20	12/04/20
PLAN B-Rebooking a new date for experiment	1 day	13/04/20	13/04/20
Root cause analysis for unsuccessful experiment (troubleshooting)	5 days	13/04/20	17/04/20
Correction of all faults in the hardware or software setup. Conducting re-checks for new experiment	6 days	17/04/20	22/04/20
DAY 2: i) Experiments at the Lake. ii) collection of results	1 day	23/04/20	23/04/20
i) Analysis of result. ii) Documentation of results	4 days	23/04/20	26/04/20
Completion of documentation and writing	77 days	25/03/20	12/06/20
Finalizing writing of Thesis.	12 days	23/04/20	04/05/20
Submission of Thesis to supervisor for final review	8 days	04/05/20	11/05/20
Final update after review and comments of supervisor on journal article	8 days	11/05/20	18/05/20
Submission of thesis.	1 day	18/05/20	18/05/20
Finalising the executive summary writing.	20 day	04/05/20	11/05/20
Submission of executive summary to supervisor for final review	7 days	11/05/20	21/05/20
Submission of executive summary	2 days	21/05/20	22/05/20
Creating presentation slides	4 days	22/05/20	25/05/20
Oral presentation practice with supervisor and colleagues	11 days	25/05/20	3/06/20
Oral presentation	1 day	5/06/20	5/06/20
Project closure	1 day	5/06/20	5/06/20

Development of a neural network-based control system for trajectory tracking, berthing and unberthing missions.

APPENDIX E: PROJECT RISK ASSESSMENT

AMC Research Job Safety Analysis and Risk Assessment															
Research Project Title:		NUMERICAL SIMULATION AND EXPERIMENTS FOR DEVELOPMENT OF TRAJECTORY TRACKING/BERTHING AND UNBERTHING SYSTEM USING NEURAL NETWORK BASED CONTROL ALGORITHM FOR A SURFACE VESSEL (BLUEFIN)													
Risk Assessment Conducted By:		Georgewill Ogunmedom		Signed:		Date:		23/03/2019		This document will be placed in the AMC Risk Assessment Register		No, don't make this available to other researchers...			
First Named Chief Investigator/Principal Investigator:				Signed:		Date:						No, don't make this available to other researchers...			
AMC Risk Owner:				Signed:		Date:						No, don't make this available to other researchers...			
Date Risk Assessment Completed:		7		If there are any changes to the scope of the work during the project a review of the risk assessment is required - this is a						Review Date(s):					
Risk Category	Describe Specific Activity/Job task or step	Identify Potential Hazard/Risks (How it could happen)	Potential Outcome (what could happen)	Initial Risk Assessment			Proposed/Existing Key Controls	Adequacy of Controls	Residual Risk Assessment			Risk Treatment Option	Risk Priority Plan	Target Risk rating	Person(s) Responsible
				Consequence	Likelihood	Inherent Risk Level			Residual Consequence	Residual Likelihood	Residual Risk Level				
Research	Mathematical modelling of the control system.	Errors in the mathematical model of the system	Incorrect simulation results which cannot be validated.	Catastrophic	Rare	High 14	Knowledge of ship dynamics and control systems will be built upon. Mathematical models will be re-checked against errors.	Strong	Moderate	Possible	High 16	Avoid the Risk	principles of ship dynamics, control systems and programming will be employed and consulted.	Mod 9	Project manager
Research	Project management	Scope becomes vague or too complex.	Project cannot continue, project is delayed.	Catastrophic	Rare	High 14	The project scope will be clearly defined and continuously reviewed to avoid complexity.	Strong	Major	Rare	Mod 10	Avoid the Risk	Project scope will be clearly defined at all stages of the project.	Mod 8	Project manager
Research	Programming and simulation	Program errors	Simulation cannot run, simulation results are incorrect and cannot be validated by experiment.	Catastrophic	Possible	High 12	The programs will be carefully checked to avoid error, competency will be ensured in using simulation softwares.	Strong	Catastrophic	Possible	High 14	Further reduce the consequences	The program will be carefully and continuously debugged and an external support maybe sought for in case of complications.	High 12	Project manager
Research	Project execution	Project product is too complex to execute	Project is delayed, Project cannot be completed.	Major	Rare	Mod 10	The feasibility of the project product will be assessed at the project start-up.	Strong	Minor	Unlikely	Low 5	Avoid the Risk	More resources (financial resources and time) will be employed as a trade-off to the project complexity.	Mod 7	Project manager
Research	Cost estimation/Scheduling	Project falls behind schedule.	Delay in completing the project, deadlines are not met.	Moderate	Unlikely	Mod 9	Appropriate time management will be employed. Project crashing will be effectively employed in the project execution.	Reasonable	Minor	Rare	Low 3	Avoid the Risk	Subsequent project activities will be crashed to cover-up for the lost time.	Mod 7	Project manager
Research	Literature review/theoretical studies	Inadequate information for theoretical studies of the project.	The project Research is based on wrong principles and assumptions, lack of competency in completing the research with correct results.	Major	Unlikely	High 13	Library and online resources will be effectively utilised in the literature review.	Strong	Insignificant	Rare	Low 1	Avoid the Risk	An extended search will be conducted to obtain valuable resources for theoretical studies.	Mod 11	Project manager
Research	Project planning	Software and hardware dependencies not properly anticipated.	Project is delayed.	Minor	Possible	Mod 8	Adequate project planning will be ensured.	Reasonable	Minor	Rare	Low 3	Avoid the Risk	The project hardware and software dependencies will be reviewed and immediate actions taken.	Mod 7	Project manager
Research	Simulation Results and analysis	Results do not meet with the expected outcome.	Results cannot be validated. Results cannot be accepted.	Catastrophic	Possible	High 14	Comprehensive risk assessment will be carried out to identify and eliminate factors impacting on the project results negatively.	Reasonable	Major	Unlikely	High 15	Avoid the Risk	The project design (modelling, programming and simulation) will be reassessed and checked against possible errors.	Mod 7	Project manager
Research	Experimental validation	Experiments do not validate simulation results	Project is unaccepted.	Catastrophic	Rare	High 14	Comprehensive risk assessment will be carried out to identify and eliminate factors impacting on the project results negatively.	Reasonable	Major	Unlikely	High 15	Avoid the Risk	Experimental setup will be reassessed, The project design (modelling, programming and simulation) will be reassessed and checked against possible errors.	Mod 6	Project manager
Financial	Cost estimation and budgeting	Wrong costing/Project exceeding budget	Project is delayed, project overshoots its budget.	Minor	Possible	Mod 8	Proper cost estimation will be conducted, cost contingencies will be put in place.	Reasonable	Minor	Rare	Low 3	Avoid the Risk	cost contingency plans (contingency funds) will be adopted.	High 13	Project manager
Financial	Cost estimation and budgeting	Inflation in price of purchase and procurement of required project materials.	Project is delayed, project overshoots its budget.	Minor	Rare	Low 3	Resources needed for the project will be purchased in an upfront-contract purchase to eliminate the effect of inflation.	Reasonable	Insignificant	Rare	Low 1	Share the Risk	cost contingency plans (contingency fund) will be adopted.	Mod 8	Project manager
Facilities	Computing of research data and information management.	Loss of data	Project is delayed.	Catastrophic	Rare	High 14	Project data will be properly backed-up at all times.	Strong	Insignificant	Rare	Low 3	Avoid the Risk	Project stakeholders will be informed and an extension of deadline will be sought for to enable recovery or reproduction of data.	Mod 5	Project manager
Facilities	Use of facilities	Equipment failure	Project cannot continue, project is delayed.	Major	Rare	Mod 10	Contingency plans will be in place against equipment failures, all equipment standard operating procedure will be consulted before the equipment use.	Reasonable	Moderate	Unlikely	Mod 9	Contingency Planning	Subsequent project activities will be crashed to cover-up for the lost time.	Mod 6	Project manager
Facilities	Use of experimental facilities	Unavailability of equipment due to high volume of users.	project is delayed.	Moderate	Likely	High 17	facility-use will be booked upfront.	Strong	Minor	Unlikely	Low 5	Avoid the Risk	Subsequent project activities will be crashed to cover-up for the lost time.	High 12	Project manager
Facilities	Use of facilities	Abnormal performance.	Incorrect results, unacceptable results, results cannot be validated.	Catastrophic	Rare	High 14	all equipment standard operating procedure will be consulted before the equipment use.	Reasonable	Moderate	Rare	Mod 6	Avoid the Risk	Experimental setup will be reassessed, The project design (modelling, programming and simulation) will be reassessed and checked against possible errors.	Mod 6	Project manager
Student/ Staff Welfare	Management of project.	Psychological and emotional stress on researcher.	project is delayed.	Major	Possible	High 14	activities and leisure will be ensured to avoid excessive workload.	Reasonable	Insignificant	Rare	Low 1	Avoid the Risk	Project workload will be reduced to allow recovery of researcher	Mod 10	Project manager
Student/ Staff Welfare	Management of project.	Health problem	project is delayed.	Catastrophic	Rare	High 14	To the best possible extent. Good health of the project team will be encouraged and maintained.	Weak	Minor	Unlikely	Low 5	Contingency Planning	Project workload will be reduced to allow recovery of researcher	Low 4	Project manager
Academic Performance or Quality	Skills	lack of skills required to complete the project	Project cannot continue, incompetency in executing project, project is delayed.	Major	Unlikely	High 15	All skills required for the project will be developed before starting the project.	Reasonable	Minor	Rare	Low 3	Avoid the Risk	All skills required for the project will be enhanced further.	Mod 6	Project manager
Academic Performance or Quality	Referencing	Plagiarism, improper referencing, copyright infringement.	Failure, project cannot continue/ Project faces legal actions.	Catastrophic	Rare	High 14	Academic honesty and integrity will be adopted throughout the project, proper referencing will be ensured through the project work, all copyrights relating to the project work will be respected	Strong	Minor	Rare	Low 3	Avoid the Risk	plagiarism and violations of copy right will be discarded completely.	Mod 6	Project manager
Academic Performance or Quality	Documentation and reporting	typographical/grammatical errors	Project work is considered sub-standard.	Moderate	Rare	Mod 6	The project reporting will be continuously proof-read to avoid errors.	Reasonable	Insignificant	Rare	Low 1	Avoid the Risk	Further checks will be carried out to eliminate all errors.	High 14	Project manager

APPENDIX F: EXPERIMENT RISK ASSESSMENT

AMC Research Job Safety Analysis and Risk Assessment															
EXPERIMENTS FOR DEVELOPMENT OF TRAJECTORY TRACKING/BERTHING AND UNBERTHING SYSTEM USING NEURAL NETWORK BASED CONTROL ALGORITHM FOR A SURFACE VESSEL (BLUEFIN)															
Research Project Title:		Georgewill Oguemedom			Signed: _____		Date: 09/07/2019		This document will be placed in the AMC			No, don't make this available to other re			
Risk Assessment Conducted By:		Georgewill Oguemedom			Signed: _____		Date: _____		This document will be placed in the AMC			No, don't make this available to other re			
First Named Chief Investigator/Principal Investigator:		Georgewill Oguemedom			Signed: _____		Date: _____		This document will be placed in the AMC			No, don't make this available to other re			
AMC Risk Owner:		Georgewill Oguemedom			Signed: _____		Date: _____		This document will be placed in the AMC			No, don't make this available to other re			
Date Risk Assessment Completed:		7			If there are any changes to the scope of the work during the project a review of the risk assessment			Review Date(s): _____		This document will be placed in the AMC			No, don't make this available to other re		
Risk Category	Describe Specific Activity/Job task or step	Identify Potential Hazards/Risks (How it could happen)	Potential Outcome (what could happen)	Initial Risk Assessment			Proposed/Existing Key Controls	Adequacy of Controls	Residual Risk Assessment			Risk Treatment Option	Risk Priority Plan	Target Risk rating	Person(s) Responsible
				Consequence	Likelihood	Inherent Risk Level			Residual Consequence	Residual Likelihood	Residual Risk Level				
Facilities	setting up the experiment	wrong hardware set-up	ship model cannot run or will show abnormal behaviour/results	Catastrophic	Unlikely	Ext 19	proper understanding of the procedures and steps for experimental setup	Reasonable	Moderate	Possible	High 16	Contingency Planning	Hardware setup procedures will be clarified in the laboratory before experiment day.	Mod 9	Project manager
Facilities	Setting up hardware equipment.	loose connections and fittings in the hardware set-up	ship model cannot run or will show abnormal behaviour/results	Catastrophic	Unlikely	Ext 19	conducting checks and re-checks on all connections and fittings	Strong	Minor	Rare	Low 3	Avoid the Risk	engineering toolbox will be made available to correct all connection and fitting faults.	Mod 8	Project manager
Facilities	transportation of experimental set-up to lake	physical damage to equipment	ship model cannot run or will show abnormal behaviour/results	Catastrophic	Unlikely	Ext 19	securing the experimental set-up properly during transportation.	Strong	Minor	Rare	Low 3	Avoid the Risk	level of damage will be ascertained and experiment rescheduled if the damage is critical to results	High 12	Project manager, technical support team.
Facilities	conducting experiments at lake	water damage to electrical/electronic parts of equipment	ship model cannot run or will show abnormal behaviour/results	Catastrophic	Unlikely	Ext 19	securing and sealing off all openings to avoid water contact during experiments	Strong	Minor	Unlikely	Low 5	Avoid the Risk	level of damage will be ascertained and experiment rescheduled if the damage is critical to results	Mod 7	Project manager
Academic Performance or Quality	conducting experiments at lake	Bad/harsh weather conditions	delay in conducting experiments, capsizing of ship model, poor gps/wireless connection.	Major	Possible	Ext 18	taking weather reports and season into consideration before booking a date for experiments.	Reasonable	Moderate	Possible	High 16	Contingency Planning	experiments will be delayed by hours or rescheduled if the harsh weather persist.	Mod 7	Project manager
Facilities	conducting experiments at lake	loss of connection between the host PC and the target PC	loss of ship model in the lake, delay in conducting experiments, experiments cannot continue.	Major	Possible	Ext 18	ensuring the prescribed trajectory allows the GPS to stay in range.	Reasonable	Moderate	Possible	High 16	Contingency Planning	backup motor-boat will be provided for recovery of experimental boat, level of damage will be ascertained and experiment rescheduled if the damage is critical to results	Mod 7	Project manager, Lab technician.
Academic Performance or Quality	conducting experiments at lake	Unexpected results and outcome	delay in conducting experiments, experiments cannot continue.	Catastrophic	Possible	Ext 22	Comprehensive checks to identify and eliminate all hardware and software factors impacting on the project results negatively.	Reasonable	Major	Possible	Ext 18	Further reduce the consequences	The experimental setup will be continuously reassessed and checked against possible errors, tools will be made available for troubleshooting of faults.	Mod 7	Project manager
Academic Performance or Quality	conducting experiments at lake	Loss of result data.	repeat of experiments	Major	Rare	Mod 10	result collection, storage and backup will be prioritised	Reasonable	Major	Unlikely	High 15	Further reduce the consequences	results will be collected and backed-up for every set of experimental run.	Mod 6	Project manager